

Byzantine-Fault-Tolerant Hierarchical Navigation for GNSS-Denied UAV Swarms: Architecture, Theoretical Analysis, and Evaluation Methodology

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ABSTRACT

Cooperative navigation of unmanned aerial vehicle (UAV) swarms under Global Navigation Satellite System (GNSS) denial has advanced rapidly, but most deployed systems implicitly assume that every peer reports honestly; adversarial-peer defence at the state-estimation layer is largely absent. We present an architecture that, to our knowledge, is the first to simultaneously address six concerns previously treated in isolation: (1) Byzantine-fault-tolerant (BFT) consensus over navigation data, (2) a hierarchical anchor/follower topology with dynamically assigned roles, (3) reputation-weighted multi-hop position relay, (4) automatic failover with sub-10-second design budget, (5) GNSS-spoofing detection through ultra-wideband (UWB) cross-verification, and (6) load balancing under progressive attrition. Based on a systematic review of approximately 90 publications (2020–2026), no prior work combines all six within a flight-grade continuous-state estimator—where *flight-grade* denotes sub-50 ms optimisation latency, sub-15 kbit/s per-UAV bandwidth, and sub-100 m circular error probable (CEP) at cruise speeds ≥ 30 m/s. The core primitive is a *Visual Proof of Observation*: cryptographically signed hashes of terrain feature descriptors that neighbouring UAVs verify by physical revisit. Reputation is updated asymmetrically ($\alpha_{\text{neg}} \geq 2\alpha_{\text{pos}}$) and mapped to factor-graph noise covariances. We derive a Byzantine safety bound and a convergence bound for naïve-Byzantine exclusion (the formal safety guarantees apply to naïve adversaries; partial mitigation against strategic adversaries is discussed but not formally proved) and quantify per-hop error propagation in closed form; under the recommended parameter set, the analysis predicts exclusion of a compromised anchor on the order of a few hundred seconds, sub-10-s follower reassignment after anchor loss, and CEP < 100 m up to three relay hops

The methods described in Sections IV-B through IV-G are the subject of pending patent applications filed with the German Patent and Trademark Office (DPMA), Aktenzeichen 10 2026 001 712.2 and a related application. This manuscript describes the technical content of those filings for the purposes of academic disclosure. *Note*: this paper presents the architecture, formal analysis, and a complete simulation methodology; empirical validation is deferred to future work where the proposed methodology will be executed in a Gazebo/ROS2-based simulation framework currently under development.

under multi-anchor fusion (the default operating mode). To support reproducible evaluation, we present a complete simulation methodology (six experiments covering CEP-vs-hops, failover timing, Byzantine exclusion, GNSS-spoofing reaction, progressive attrition, and ablation) together with the analytical predictions each is designed to validate. Empirical execution of the methodology in a Gazebo/ROS2 framework, hardware-in-the-loop validation, and head-to-head comparison against the original SwarmRaft implementation are deferred to future work.

Index Terms—UAV swarm, Byzantine fault tolerance, GNSS-denied navigation, factor graph, reputation

I. INTRODUCTION

UNMANNED aerial vehicle (UAV) swarms are increasingly deployed in contested environments where GNSS signals are unavailable, degraded, or deliberately denied. Recent operational experience in military and civil domains has underscored two coupled vulnerabilities: GNSS-dependent navigation collapses under electronic warfare, and cooperative navigation schemes that share position information between vehicles assume that every participant reports honestly. A single compromised UAV—captured, sensor-faulted, or infected with malicious firmware—that transmits falsified odometry or absolute-position estimates can corrupt the navigation solution of the entire swarm without detection.

Three research streams have independently matured in the past five years. First, cooperative GNSS-denied navigation has produced flight-grade systems such as the Distributed PNT Chain [5], Omni-swarm [6], Kimera-Multi [8], and Swarm-SLAM [9], and anchor-free multi-sensor fusion schemes such as DTVIRM-Swarm [43] have added magnetic and monocular cues. Second, Byzantine-fault-tolerant (BFT) protocols for robot swarms have evolved from abstract consensus [13] to deployed token economies [14] and blockchain-secured cluster hierarchies [44]. Third, reputation and trust frameworks for multi-agent systems have converged on asymmetric update rules and physical-layer observations [20]–[22]. Yet these streams have not merged at the flight-control layer. The two 2025 systems closest to a unified architecture—

SwarmRaft [2] and the Byzantine Distributed GPS Detection (BDGD) algorithm [3]—each cover at most three of the design dimensions required for a deployable solution; an earlier and largely complementary system, the Byzantine Follow-The-Leader (BFTL) formalism of Castelló Ferrer *et al.* [16], covers four, but operates on discretised maps rather than a continuous flight-grade state estimator.

This paper introduces a navigation architecture that simultaneously addresses six pillars: (i) BFT consensus over shared state, (ii) a hierarchical anchor/follower topology with roles assigned dynamically in flight, (iii) reputation-weighted multi-hop position relay, (iv) automatic failover within 10 seconds of anchor loss, (v) GNSS-spoofing detection via UWB cross-verification, and (vi) load balancing across anchors under progressive attrition. Based on a systematic review of approximately 90 publications spanning 2020–2026 across Byzantine-fault-tolerant swarms, GNSS-denied cooperative navigation, multi-robot reputation systems, hierarchical swarm control, GNSS-spoofing detection, and multi-hop position relay (see Section II footnote for review methodology), we find, to our knowledge, no published system integrating more than four of these pillars, and none doing so within a continuous-state flight-grade factor-graph estimator. The architecture rests on a new primitive, the *Visual Proof of Observation* (PoO): each terrain-relative navigation (TRN) fix produces a cryptographically signed record containing a hash of terrain feature descriptors, which neighbouring UAVs verify by comparing their own descriptors when they later fly over the claimed region. Because feature descriptors are physically bound to terrain, illumination, and the learned extractor network, they cannot be fabricated by a vehicle that has not actually observed the location, distinguishing this mechanism from signal-level or purely statistical trust.

Our contributions are as follows. *First*, we formulate a visual PoO mechanism that gives distributed navigation a terrain-anchored verification primitive (physically bound to terrain, illumination, and the learned feature extractor; unforgeable in the practical sense under the threat model of Section III, although not under the formal cryptographic UF-CMA definition), and we show how to integrate signed observations into an incremental smoothing and mapping (iSAM2) factor graph as reputation-weighted priors alongside UWB ranging factors. *Second*, we define an asymmetric reputation rule ($\alpha_{\text{neg}} \geq 2\alpha_{\text{pos}}$, preferably 3:1) that excludes Byzantine participants faster than honest peers accumulate trust while tolerating transient verification failures from environmental degradation. *Third*, we introduce a dynamic anchor-score assignment combining reputation, TRN confidence, imagery coverage, and load, together with a “request–offer–select” failover protocol with bounded recovery time. *Fourth*, we derive a safety condition for quorum-based exclusion, expose its limits when the quorum parameter M is fixed, give a closed-form expected-time bound for naïve-Byzantine exclusion, and present a closed-form per-hop error-growth model. *Fifth*, we present a complete simulation evaluation methodology (six experiments covering CEP-vs-hops, failover timing, Byzantine detection

and exclusion, GNSS-spoofing reaction, progressive attrition, and ablation), each paired with the analytical prediction it is designed to validate; empirical execution of this methodology is deferred to future work in a Gazebo/ROS2 framework currently under construction.

The remainder of the paper is organised as follows. Section II surveys related work. Section III formalises the system and threat model. Section IV details the proposed approach in seven subsections. Section V presents the theoretical analysis. Section VI presents the proposed evaluation methodology with the analytical prediction each experiment is designed to test. Sections VII, VIII, and IX report three abstract Monte Carlo validation studies of the asymmetric reputation rule, the convergence bound, and Byzantine-fraction resilience respectively. Section X discusses limitations and deployment considerations. Section XI concludes.

II. RELATED WORK

We organise prior art into five domains and summarise design-space coverage of the most relevant systems in Table I.¹

A. Byzantine Fault Tolerance in UAV Swarms

The Byzantine fault model originates with Lamport, Shostak, and Pease [12], who established the foundational $n \geq 3f + 1$ requirement for consensus among n peers in the presence of f arbitrarily-faulty participants. Practical implementations have evolved through two main families described below.

Raft-family adaptations. The original Raft consensus algorithm [1] provides crash-fault tolerance with leader-based replication; UAV-specific extensions add Byzantine resilience on top of this substrate. SwarmRaft [2] performs median-based correction of spoofed GNSS positions among leader-collected peer reports and includes a Byzantine-resilient evaluation step on top of a crash-fault Raft core. Its design choice is not presented as a limitation by the authors, but the system lacks persistent reputation across rounds, weights all peers equally within a round, and retains a single elected leader. Zhang *et al.* [36] overlay Proof-of-Authority on Raft for UAV routing, not localisation. Zuo *et al.* [37] provide Raft-inspired leader election with explicit follower-loss failover but no reputation.

PBFT variants. Practical Byzantine Fault Tolerance, originally introduced by Castro and Liskov [30], has spawned a family of UAV-specific extensions. LAP-BFT [31] provides lightweight asynchronous BFT with vector commitments over

¹Review methodology: candidate publications were identified by keyword search in Google Scholar, arXiv, IEEE Xplore, and ACM Digital Library between February and April 2026 using combinations of $\{\textit{Byzantine}, \textit{BFT}, \textit{fault-tolerant}, \textit{reputation}, \textit{trust}\} \times \{\textit{UAV}, \textit{drone}, \textit{swarm}, \textit{multi-robot}\} \times \{\textit{navigation}, \textit{localisation}, \textit{SLAM}, \textit{GNSS}, \textit{GPS}\}$; inclusion criteria were peer-reviewed conference or journal venue (or arXiv preprint with an identifiable author affiliation), publication date 2020–2026, and at least partial coverage of one of the six pillars enumerated in Section I. Of approximately 90 candidate publications screened, the systems explicitly compared in Table I are those that cover three or more pillars. The “to our knowledge” hedge in the abstract and conclusion reflects the irreducible incompleteness of any keyword-based search.

50 UAVs; DTPBFT [32] extends PBFT with a dynamic trust mechanism for UAV swarms; ACBFT [33] is an adaptive chained Byzantine-fault-tolerant consensus protocol for UAV ad-hoc networks; DPoS-PBFT [34] hybridises delegated proof-of-stake with PBFT finality. The concurrent reputation-enhanced PBFT variant RePA [35] adds trust scoring to defend against node-capture attacks at the network layer. Concurrent work on parallelised BFT for swarm robots (PTEE-BFT [67]) introduces a parallel-execution Byzantine consensus protocol for blockchain-secured swarms, addressing throughput rather than navigation. None of these targets localisation. The recent COUAVChain [4], an RB-HotStuff-based variant building on the HotStuff BFT protocol [64], couples dynamic reputation, Bayesian inference, and hierarchical sharding for denied environments—architecturally closest to ours on the consensus side—but does not close the loop with spoofing detection or multi-hop position relay.

Blockchain-based swarm reputation. The IRIDIA/ULB line established Byzantine resilience for physical robot swarms [13], [14], [17], [18]. Strobel *et al.* [14] demonstrated a token economy on Pi-puck swarms that excludes adversaries autonomously. Castelló Ferrer *et al.* [15] extended the framework with secure multi-party computation for confidential missions. Among these, the BFTL system [16] is the closest prior navigation-aware work: it formalises the Byzantine Follow-The-Leader problem, publishes cryptographically weighted routes to a blockchain, and proves completion bounds under Byzantine leaders and followers (including a form of leader-failure tolerance arising from the completion guarantees). BFTL covers BFT, hierarchy, reputation, and position relay but operates over discretised graph-structured environments (with routes committed to blockchain) rather than a continuous probabilistic factor-graph state estimator. The blockchain settlement layer used in the published BFTL demonstration has route-commitment finality on the order of seconds per round [13] — a property of the chosen settlement layer rather than of the BFTL formalism itself, but the published instantiation as deployed is consequently incompatible with 50-ms control cycles; a future BFTL implementation over a faster settlement layer could in principle close this gap, and the BFT design choices that distinguish our work from BFTL are independent of the settlement-layer latency question (continuous-state factor-graph estimation, dynamic failover, and load balancing in particular). Alsamhi *et al.* [44] link blockchain trust to cluster-head hierarchy with energy awareness but without multi-hop relay or spoofing defence. Jensen *et al.* [46] and De Campos *et al.* [45] explored earlier blockchain-UAV integrations.

Distributed resilient estimation. The W-MSR track originating with LeBlanc *et al.* [24], with later extensions to trusted-node settings by Mitra *et al.* [25], provides formal scalar-state guarantees but lacks flight-grade sensor-fusion detail.

B. GNSS-Denied Cooperative Navigation

The Distributed PNT Chain [5] from GTRI factors a centralised Kalman filter over swarm poses into an undirected acyclic Bayes graph, matching centralised accuracy at $\sim 22\%$ of the compute cost. It does not defend against peers that falsify odometry—a spoofed node propagates unchecked through the factor graph. Omni-swarm [6] and its successor D²SLAM [7] are canonical decentralised UWB+VIO systems; CCM-SLAM [10], COVINS [11], Kimera-Multi [8], and Swarm-SLAM [9] are the leading collaborative VIO/SLAM frameworks, evaluated on swarms of fewer than 20 agents. None defends against falsified loop closures or odometry; statistical outlier rejection (GNC, robust kernels) is routine, but deliberate adversarial behaviour is out of scope. Moroncelli *et al.* [19] extended Swarm-SLAM with blockchain-based geometric Byzantine detection—the closest prior coupling of reputation with SLAM, although operating on ground robots and without hierarchical failover.

The “Collaborative Integrated Navigation” scheme of Zhang *et al.* [41] models uncertainty propagation from a leader to followers via UWB ranging but employs only two hierarchy levels with a statically assigned leader, no reputation, and no failover. The air-ground scheme of Yue *et al.* [42] uses high-altitude UAVs as fixed-role mobile UWB anchors without dynamic election, reputation propagation, or failover. DTVIRM-Swarm [43] is the most comprehensive honest-peer anchor-free fusion to date (MIMU + magnetic + monocular + UWB with χ^2 outlier filtering), but assumes statistical rather than adversarial outliers.

C. Reputation Systems for Multi-Robot Systems

Yemini *et al.* [20] injected stochastic physical-layer trust into consensus, breaking the classical $< 50\%$ malicious bound. Ballotta *et al.* [21] added time-decaying confidence for tunable convergence. Mallmann-Trenn, Cavorsi, and Gil [22] extended this idea to flocking through “crowd vetting” of trust opinions. FLTrust [23] is the federated-learning analogue. Closer to the adversarial cooperative-localisation setting, Fang *et al.* [71] present an information-theoretic analysis of trustworthy UAV cooperative localisation against malicious peers; their treatment is single-hop and does not propose a multi-hop reputation-weighted noise-covariance propagation model, which remains the distinguishing mechanism in our architecture. Most of this work operates at the decision or aggregation layer, not the flight-control estimator.

D. GNSS-Spoofing Detection in Swarms

GPS spoofing of civilian UAVs was first demonstrated experimentally by Kerns *et al.* [65]. Mykytyn *et al.* [26] subsequently introduced UWB cross-verification against the Spherical Law of Cosines for GPS-spoofing detection in UAV swarms, demonstrated for small-swarm settings on consumer-grade hardware; their evaluation does not extend to swarm-wide multi-hop relay or hierarchical anchor/follower failover. Single-UAV anti-spoofing schemes such as Chimera authentication [29] and SemperFi [28] address signal-level integrity

at one receiver but do not extend to swarm-wide cross-verification.

E. Hierarchical Swarms, Position Relay, and Load Balancing

The formation-control hierarchy of Chen *et al.* [38] scales but does not consider adversarial peers. Multi-hop cooperative localisation has been studied extensively in the honest-peer setting [39], [40]. For benign (non-adversarial) localization dropout, Horyna *et al.* [68] show that UAV swarms adapt better to anchor localization loss than single robots by leveraging relative inter-agent information; this is complementary to the proposed system, which addresses adversarial Byzantine anchor behavior rather than benign sensor dropout, but to the best of our knowledge no published analysis addresses adversarial multi-hop UWB error propagation; existing multi-hop schemes realise relay without *reputation-weighted* noise propagation, which is the distinguishing mechanism in our architecture. Recent learned-coordination schemes target adjacent problems: RALLY [47] is an LLM-driven role-adaptive yoked-navigation framework that outperforms multi-agent reinforcement-learning and LLM baselines on agentic UAV swarms, while CI-HRL [48] applies decentralised consensus-inference hierarchical reinforcement learning to multi-constrained UAV pursuit-evasion. Neither incorporates Byzantine resilience or trust-weighted state estimation. For load balancing, CLAC-CBBA [49] combines centrality-based cluster reconfiguration with load-aware self-regulation for heterogeneous swarms, but operates at the task-allocation layer without reputation weighting or Byzantine exclusion. For failover, Wu *et al.* [50] propose a QMIX/DQN-based multi-agent scheme that redeploys UAVs after mission-destruction events, without BFT, spoofing detection, or multi-hop relay. Finally, the concurrent Swarm Oracle work [51] demonstrates a blockchain-based BFT protocol on 12 physical robots at the theoretical $f=n/3$ limit, integrating a token-based reputation system that penalises both unreliable and inactive robots—but addresses collective perception for blockchain oracles rather than flight-grade navigation under GNSS denial.

III. SYSTEM MODEL

Swarm composition. We consider a swarm $\mathcal{N} = \{v_1, \dots, v_n\}$ operating in a region without GNSS (or with partially spoofed GNSS). Each vehicle carries an IMU, a UWB ranging radio (accuracy ± 0.1 m, range 200–500 m), a communication transceiver, a computing unit with a secure key store, and a barometric altimeter. A subset $\mathcal{A}_{\text{cap}}$ ($|\mathcal{A}_{\text{cap}}|/n \approx 0.025\text{--}0.05$, i.e., 5–10 TRN-capable anchors per 200 UAVs) additionally carries a downward-facing camera and pre-stored satellite imagery, enabling terrain-relative navigation (TRN) with a learned feature extractor (e.g., SuperPoint [53]) and an incremental factor-graph optimiser (GTSAM/iSAM2 [52]). At any time each UAV is in one of three roles—*active anchor*, *follower*, or *standby anchor*—assigned by an anchor-score election (Section IV-E) and changeable in flight.

Threat model. The adversary can (a) capture up to f UAVs and replace their firmware while retaining valid cryptographic

TABLE I

DESIGN-SPACE COVERAGE OF CLOSEST PRIOR ART. ✓ INDICATES PUBLISHED SUPPORT; – INDICATES ABSENT; (∼) INDICATES PARTIAL OR MECHANISM-EQUIVALENT SUPPORT. COLUMNS: *BFT*, BYZANTINE FAULT TOLERANCE; *Hier.*, ANCHOR/FOLLOWER HIERARCHY; *Repu.*, PERSISTENT REPUTATION; *Fail.*, AUTOMATIC FAILOVER; *Spoof.*, GNSS-SPOOFING DETECTION; *Relay*, MULTI-HOP POSITION RELAY. SWARMRAFT’S (∼) IN THE *Fail.* COLUMN REFLECTS RAFT LEADER RE-ELECTION, WHICH HANDLES LEADER CRASH BUT DOES NOT REDISTRIBUTE FOLLOWERS OR REBALANCE LOAD UPON LEADER LOSS; ITS (∼) IN *BFT* REFLECTS A BYZANTINE-RESILIENT EVALUATION LAYER OVER A CRASH-FAULT RAFT CORE. BFTL’S (∼) IN *Fail.* REFLECTS THE FORMAL ROUTE-COMPLETION GUARANTEES UNDER BYZANTINE LEADER LOSS PROVED IN THE BFTL PAPER, WHICH CONSTITUTE A FORM OF LEADER-FAILURE TOLERANCE WITHOUT AN EXPLICIT FOLLOWER-REDISTRIBUTION PROTOCOL. THE *Repu.* COLUMN CREDITS BFTL FOR BLOCKCHAIN-WEIGHTED ROUTE COMMITMENTS (THE BFTL MECHANISM), WHICH DIFFERS FROM OUR REPUTATION-WEIGHTED NOISE-COVARIANCE MODULATION OF A CONTINUOUS STATE ESTIMATOR; THE COLUMN ENTRY RECORDS THAT A REPUTATION-FLAVOURED TRUST MECHANISM IS PRESENT, NOT THAT THE MECHANISMS ARE EQUIVALENT.

System	BFT	Hier.	Repu.	Fail.	Spoof.	Relay
SwarmRaft [2]	(∼)	–	–	(∼)	✓	–
BDGD [3]	✓	–	✓	–	✓	–
COUAVChain [4]	✓	✓	✓	–	–	–
BFTL [16]	✓	✓	✓	(∼)	–	✓
GTRI PNT Chain [5]	–	–	–	–	–	✓
Kimera-Multi [8]	–	–	–	–	–	✓
Omni-swarm [6]	–	–	–	–	–	✓
Swarm-SLAM [9]	–	–	–	–	–	✓
Moroncelli [19]	✓	–	✓	–	–	✓
Strobel [14]	✓	–	✓	–	–	–
Zhang <i>et al.</i> [41]	–	✓	–	–	–	(∼)
Yue <i>et al.</i> [42]	–	✓	–	–	–	✓
Mykytyn [26]	–	–	–	–	✓	–
This work	✓	✓	✓	✓	✓	✓

keys; (b) inject localised GNSS spoofing affecting a subset of the swarm [62]; (c) perform mass GNSS spoofing; (d) selectively jam links. The adversary cannot forge terrain descriptors for unobserved locations, cannot forge Ed25519 signatures without private keys, and cannot introduce Sybil identities [63] absent an out-of-band enrolment step. We distinguish *naïve* Byzantine peers (always falsify) from *strategic* peers (honest when observed, falsifying at critical moments); guarantees differ between these cases (Section V). Safety of quorum-based permanent exclusion requires $f \leq M - 1$, where M is the configurable quorum parameter (analysed in Section V).

Budgets and requirements. Per-UAV budget: <15 kbit/s bandwidth, <50 ms factor-graph optimisation on a ~ 100 -TOPS embedded accelerator. No external infrastructure (ground station, cloud, or blockchain validator). The swarm must maintain CEP <100 m for NATO-grade ISR, with continuous navigation under anchor destruction, Byzantine compromise, localised GNSS spoofing, and progressive attrition over 1–3 hour missions.

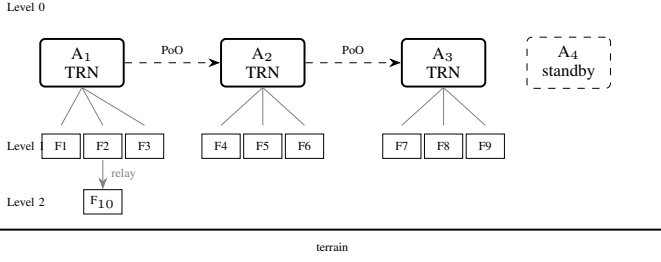


Fig. 1. Hierarchical architecture. Active anchors perform mutual Visual Proof of Observation (PoO) between one another and broadcast signed positions over UWB. Followers (Level 1) receive reputation-weighted position directly; deeper followers (Level 2+) receive relayed positions with per-hop noise growth. Standby anchors remain TRN-capable but are not currently elected.

IV. PROPOSED APPROACH

We address the six pillars across seven subsections: Sections IV-B and IV-D establish BFT (Pillar 1); Section IV-C defines reputation (Pillar 3); Section IV-E covers the anchor/follower hierarchy (Pillar 2) and reputation-weighted multi-hop relay (Pillar 3); Section IV-F details automatic failover (Pillar 4); and Section IV-G addresses GNSS-spoofing detection (Pillar 5) and load balancing (Pillar 6). All hyperparameters introduced across these subsections are consolidated in Table II.

A. System Architecture Overview

Each UAV runs a common navigation module with four components: a *navigation core* performing visual-inertial odometry (VIO) at 50 Hz and TRN at 0.1 Hz (TRN only on anchors), fused in a local factor graph; an *observation recorder* that produces Signed Observations on each TRN fix; a *mesh protocol handler* managing signed-observation exchange, reputation tracking, and Byzantine rejection; and an *autopilot adapter* layer. The hierarchical topology overlays an anchor/follower structure on this substrate (Fig. 1): anchors perform PoO verification mutually, while followers receive reputation-weighted position via UWB relay with up to $K_{\max}$ hops.

B. Visual Proof of Observation

On each TRN fix at position $\mathbf{p}_i \in \mathbb{R}^3$ at time t_1 , UAV v_i extracts $N = 50$ top-scoring feature descriptors $\mathbf{f}_i \in \mathbb{R}^{N \times D}$ ($D=256$ for SuperPoint). $N = 50$ is a working default: SuperPoint typically yields 200–500 keypoints per image [53]; retaining the top-50 by score balances descriptor packet size (50 descriptors at 256 bytes each = 12.8 kB) against matching distinctiveness. Sensitivity to N is deferred to Track 2. The descriptor set forms the hash

$$H_i = \text{SHA-256}(\text{sort}_{\downarrow}(\mathbf{f}_i[0:N])). \quad (1)$$

It constructs the Signed Observation

$$\text{OBS}_i = (t_1, \mathbf{p}_i, H_i, c_i, \text{ID}_i, \sigma_i), \quad (2)$$

where c_i is the RANSAC inlier ratio and

$$\sigma_i = \text{Ed25519.sign}(\text{sk}_i, t_1 \|\mathbf{p}_i\| H_i \|c_i\| \text{ID}_i). \quad (3)$$

The base record (Mode B) totals approximately 180–200 bytes: t_1 (timestamp, 8 B) + $\mathbf{p}_i$ (position [lat, lon, alt] as three doubles, 24 B) + H_i (SHA-256 hash, 32 B) + c_i (RANSAC inlier ratio as a float, 4 B) + ID_i (UAV identifier, 8 B) + σ_i (Ed25519 signature, 64 B) = 140 B core payload, plus message framing (type byte, length, sequence number, CRC, padding) accounting for the remaining 40–60 B. A *Mode A* variant appends a compact feature set of $K = 20$ descriptors. The value $K = 20$ is the minimum that keeps Mode-A packet overhead within budget: $20 \times 256 = 5120$ bytes = 5.12 kilobytes (KiB) ≈ 5.12 kB (binary convention) = 5.120 kB (SI convention, 10^3 -byte base), within the UAV datalink budget assumed in Section II, bringing total size to ≈ 5.30 kB (5120 bytes of descriptors plus the ~ 180 -byte base record). Ed25519 [54] signature computation is well within sub-millisecond on modern processors at the cycle counts reported by [54] (84932 cycles for signing); typical ARM Cortex-class embedded targets fall in the same regime, with cycle-exact timings to be characterised on the target board in Track 2.

On receiving OBS_i , a neighbour v_j first verifies σ_i using pk_i . If the signature is invalid, reputation R_i is decreased by a signature penalty $\alpha_{\text{sig}} = 0.30$, a working default equal to $2 \times \alpha_{\text{neg}} = 0.30$ so a single signature mismatch has the same net reputation impact as two consecutive failed verifications. At a later time $t_2 > t_1$, when v_j flies within verification radius $r_{\text{verify}} = 500$ m of $\mathbf{p}_i$, it extracts local features $\mathbf{f}_j$ and computes a verification score. The value $r_{\text{verify}} = 500$ m matches the upper end of the UWB operational range (200–500 m at the datalink power levels assumed in Section II), ensuring that any follower within UWB communication range can request verification without requiring out-of-range RF links. In *Mode A* (compact feature set), for each transmitted descriptor, the nearest neighbour in $\mathbf{f}_j$ is accepted if its L_2 distance is below $\tau_{\text{match}} = 0.7$ (Lowe’s ratio test threshold [56]: a threshold of 0.7 is the standard lower bound of the accepted range 0.7–0.8 for filtering ambiguous matches, used here with SuperPoint’s 256-dimensional L_2 -normalised descriptors [53]); the score V is the ratio of accepted matches to transmitted descriptors, and VERIFIED is declared when $V > T_{\text{verify}} = 0.3$. The threshold $T_{\text{verify}} = 0.3$ corresponds to 6/20 Mode-A descriptors (or 6/ N hash matches in Mode B) successfully passing the $\tau_{\text{match}} = 0.7$ Lowe-ratio test. Published cross-view SuperPoint matching studies report inlier ratios of 60–90% under clean line-of-sight conditions, falling to 20–40% under moderate viewpoint or illumination drift. Setting $T_{\text{verify}} = 0.3$ places the VERIFIED boundary at the lower end of the degraded-but-viable regime, so legitimate verifications continue to pass during moderate environmental noise while complete-failure cases (match ratios below $\sim 20\%$, characteristic of wrong-terrain or fabricated-hash submissions from a peer that has not physically observed the claimed region) are reliably rejected. The threshold interacts with τ_{match} and N, K : tightening any of the three has the same qualitative effect of raising the false-reject rate against environmentally-degraded honest verifications. Sensitivity over $T_{\text{verify}} \in \{0.2, 0.3, 0.4\}$

is deferred to Track 2. Because $T_{\text{verify}} = 0.3$ sits at the lower end of the degraded-but-viable regime (20–40% inlier ratios), honest peers operating under moderate environmental degradation can routinely produce match scores near the exclusion boundary; combined with the asymmetric reputation rule (UNVERIFIED penalty $\alpha_{\text{neg}} = 0.15$), this implies that as few as $\lceil (R_{\text{init}} - T_{\text{reject}}) / (\alpha_{\text{neg}} R_k) \rceil = 2$ consecutive boundary-case UNVERIFIED verdicts under $R_k \approx 1$ would drive an honest peer’s reputation to T_{reject} . A formal false-acceptance / false-rejection (FAR/FRR) characterisation under realistic illumination, viewpoint, and seasonal terrain variation is required to confirm the threshold choice and is part of the Track 2 calibration plan; the corresponding empirical false-exclusion rate is also reported in Limitations (Section X). A *Mode B* variant (hash-only, zero-bandwidth verification) compares recomputed hashes directly; Mode B is suited to severely bandwidth-constrained tactical deployments in which flight conditions permit reproducible descriptor extraction, and requires additional calibration against the observation geometry (see Section X).

The unforgeability rationale is physical: terrain feature descriptors are functions of the surface geometry, illumination, viewing angle, and the learned extractor. A captured UAV cannot fabricate descriptors for a location it has not physically observed without either possessing the imagery database and extractor (which still requires knowledge of the exact atmospheric and illumination conditions at t_1) or compromising honest peers before they verify—neither of which is feasible under our threat model. PoO is related to but distinct from the proof-of-location (PoL) literature: PoL systems such as VRChain [69] (blockchain-anchored visual homing) and BFT-PoLoc [70] (Byzantine-fortified trigonometric proof of location from Internet round-trip delays) verify a device’s *current* claimed position via either ledger consensus or signal-timing measurements. Visual PoO instead verifies a *past* observation through a neighbour’s later physical revisit, with evidence bound to terrain geometry via the learned extractor rather than to network timing or co-located witnesses.

C. Asymmetric Reputation System

Each UAV maintains a reputation vector $\mathbf{R} = (R_1, \dots, R_n)$ with $R_j \in [0, 1]$ initialised to $R_{\text{init}} = 0.5$ (neutral prior). The initial reputation is set to the midpoint of the $[0, 1]$ range, representing maximum uncertainty about an unknown peer: neither trusted nor distrusted. This is the standard uninformative prior for reputation systems [57]. Updates on verification events are asymmetric:

$$\text{VERIFIED: } R_j \leftarrow \text{clamp}(R_j + \alpha_{\text{pos}} R_k, 0, 1), \quad (4)$$

$$\text{UNVERIFIED: } R_j \leftarrow \text{clamp}(R_j - \alpha_{\text{neg}} R_k, 0, 1), \quad (5)$$

$$\text{INVALID SIG: } R_j \leftarrow \text{clamp}(R_j - \alpha_{\text{sig}}, 0, 1), \quad (6)$$

where R_k is the reputation of the verifier. We require $\alpha_{\text{neg}} \geq 2\alpha_{\text{pos}}$, preferably $\alpha_{\text{neg}} \geq 3\alpha_{\text{pos}}$; the default values are $\alpha_{\text{pos}} = 0.05$, $\alpha_{\text{neg}} = 0.15$. The 3:1 ratio ($\alpha_{\text{neg}}/\alpha_{\text{pos}}$) is set in Section V-B; the absolute level is anchored by the convergence

bound of Proposition 3: to achieve $T_{\text{ind}} \approx 170$ s with $R_{\text{init}} = 0.5$, $T_{\text{reject}} = 0.2$, and $\nu_{\text{verify}}^{-1} \approx 45$ s, the approximation requires $\alpha_{\text{neg}} \cdot R_{\text{honest}} \cdot \nu_{\text{verify}} \geq (R_{\text{init}} - T_{\text{reject}}) / T_{\text{ind}} \approx 0.30 / 170 \approx 0.0018$ per second, or $\alpha_{\text{neg}} \geq 0.0018 \times 45 \approx 0.08$ per verification event; $\alpha_{\text{neg}} = 0.15$ satisfies this with a $\approx 2\times$ margin. The value $\alpha_{\text{pos}} = 0.05$ follows from the 3:1 ratio. Under this asymmetry, three failed verifications outweigh nine successful ones ($3 \cdot 0.15 = 9 \cdot 0.05 = 0.45$), guaranteeing that distrust accumulates faster than trust while retaining a margin of 2–8 consecutive environment-induced failures before an honest peer is falsely excluded (2 pure-burst events from $R_{\text{init}} = 0.5$ at $R_k = 1.0$; up to 8 interleaved VERIFIED+UNVERIFIED rounds from $R_j = 1.0$; derivation follows). Let R_j be the *target* peer’s current reputation and R_k the *verifier*’s reputation weight. Each UNVERIFIED event decrements R_j by $\alpha_{\text{neg}} \times R_k$; each VERIFIED event increments R_j by α_{pos} . For pure-UNVERIFIED bursts (worst case), events needed to drive R_j to $T_{\text{reject}} = 0.2$ are $\lceil (R_j - T_{\text{reject}}) / (\alpha_{\text{neg}} \times R_k) \rceil$. At $R_k = 1.0$ and $R_j = R_{\text{init}} = 0.5$: $\lceil 0.3/0.15 \rceil = 2$; at $R_j = 1.0$: $\lceil 0.8/0.15 \rceil = 6$. For interleaved sequences (one VERIFIED plus one UNVERIFIED per round), net loss per round is $\alpha_{\text{neg}} \times R_k - \alpha_{\text{pos}}$; at $R_k = 1.0$, $0.15 - 0.05 = 0.10$, giving 3 rounds from $R_j = 0.5$ to 8 rounds from $R_j = 1.0$. Overall margin: 2 (pure burst from R_{init}) to 8 (interleaved from $R_j = 1.0$) consecutive failures.

To handle dynamic environments, we apply a temporal decay

$$V_{\text{weighted}} = V \cdot \exp(-\lambda \Delta t), \quad (7)$$

with $\lambda = 10^{-3} \text{ s}^{-1}$ (half-life ≈ 11.5 min). The decay constant is chosen so that an observation becomes effectively stale ($< 5\%$ residual weight) after ≈ 3 half-lives = 34.5 min, approximately one-tenth of the 2-hour mission (the one-tenth fraction is an operationally discretionary design goal: it ensures that any single mission phase does not permanently dominate the reputation estimate, while still allowing short-horizon events to decay within one mission); a temporary gap of one revisit cycle ($\nu_{\text{verify}}^{-1} \approx 45$ s) causes only $\approx 4.4\%$ decay per missed event ($e^{-10^{-3} \times 45} \approx 0.956$). These are working defaults pending Track 2 sensitivity analysis. Observations older than $T_{\text{max_verify}} = 3600$ s (1 hour) are considered stale. This cutoff is set to half the 2-hour planned mission length, ensuring that observations from the first half of a mission are discarded before the second half begins. For shorter or longer missions the cutoff should be scaled proportionally.

D. Reputation-Weighted Distributed Factor Graph

Each UAV maintains a local factor graph $\mathcal{G}_i$ whose maximum-a-posteriori estimate is

$$\mathbf{X}_i^* = \arg \min_{\mathbf{X}} \sum_k \|h_k(\mathbf{X}) - \mathbf{z}_k\|_{\Sigma_k}^2. \quad (8)$$

Three classes of factors are instantiated:

- *Local factors* (IMU preintegration [61], VIO, TRN-absolute on anchors, barometric altitude), standard in the art.

- *Ranging factors* for each UWB measurement d_{ij} with $h_{\text{range}}(\mathbf{X}_i, \mathbf{X}_j) = \|\mathbf{p}_i - \mathbf{p}_j\|_2$, $\sigma_{\text{range}} = 0.1$ m.
- *Reputation-weighted priors* from the Distilled State of neighbour j , where the *Distilled State* of UAV j is the per-UAV broadcast tuple $\text{DS}_j = (t, \mathbf{p}_j, \mathbf{v}_j, \text{cov}(\mathbf{p}_j), R_j, \text{ID}_j, \sigma_j^{\text{Ed25519}})$ comprising timestamp, position, velocity, position-covariance summary, current reputation, identifier, and Ed25519 signature; the byte-level payload composition and bandwidth accounting are given in Section V-D. The Distilled State is the central inter-UAV communication primitive of the architecture and is consumed by the reputation-prior factor here, by the failover watchdog (Section IV-F, phase F1), and by the load-balancing broadcast (Section IV-G). The reputation-prior factor uses the position $\mathbf{p}_j$ and covariance $\text{cov}(\mathbf{p}_j)$ of DS_j , scaled by R_j , with noise

$$\sigma_j = \sigma_{\text{base}} / (R_j + \varepsilon), \quad (9)$$

$\sigma_{\text{base}} = 50$ m, $\varepsilon = 0.01$ (dimensionless numerical regularisation; chosen two orders of magnitude below $\sigma_{\text{base}} = 50$ m so $\varepsilon/\sigma_{\text{base}} = 0.02\%$, ensuring negligible influence on the computed covariance). Covariance $\Sigma_j = \text{diag}(\sigma_j^2, \sigma_j^2, (\sigma_j/2)^2)$ for (lat, lon, alt); the halved altitude term reflects a standard anisotropy assumption that UAV barometric altimeters provide tighter altitude bounds than UWB-derived lateral uncertainty. Experiment 1 will characterise whether the actual altitude/lateral ratio differs from 1:2. Sensitivity to σ_{base} : the analytical CEP bound in Proposition 3 scales linearly with σ_{base} ; doubling to 100 m would require approximately $2\times$ as many anchor observations to converge to the 100 m CEP pass criterion. A full sensitivity sweep over $\sigma_{\text{base}} \in \{25, 50, 75, 100\}$ m is planned in Track 2.

A UAV with $R = 1.0$ contributes $\sigma \approx 49.5$ m, versus $\sigma \approx 238$ m at $R = 0.2$ —approximately a $5\times$ weight ratio in precision and $\sim 23\times$ in variance. Optimisation uses iSAM2 with the Bayes-tree representation; the <50 ms per-update latency on an embedded accelerator is a design target consistent with published GTSAM/iSAM2 benchmarks [52]. The <50 ms budget covers the iSAM2 optimisation step only; the full PoO pipeline (camera capture ~ 10 ms, SuperPoint inference ~ 20 – 40 ms on a 50–100 TOPS-class accelerator at 480×640 resolution per published embedded benchmarks [53], top- N sort <1 ms, SHA-256 hash <1 ms, and Ed25519 signing <1 ms at the 84932-cycle figure from [54] on an ARM Cortex-class core) is constrained by the 0.1 Hz TRN fix rate ($=10$ s budget) rather than by the 50 ms factor-graph cycle, so SuperPoint inference timing is not on the critical sub-50 ms path. Per-board characterisation of the PoO pipeline on the specific target accelerator is a Track 2 calibration item; the 50–100 TOPS class is named here as a representative envelope rather than a specific SKU.

A UWB consistency check provides a secondary, non-visual channel: if UAV j reports position $\mathbf{p}_j^{\text{rep}}$ but UWB ranging d_{ij} yields $\|\mathbf{p}_i - \mathbf{p}_j^{\text{rep}}\| - d_{ij} > \delta_{\text{consist}} = 50$ m, reputation

decreases by $\alpha_{\text{range}} = 0.10$. The threshold $\delta_{\text{consist}} = 50$ m matches $\sigma_{\text{base}} = 50$ m (the declared prior noise), so a consistency check flags deviations beyond one prior standard deviation. The coincidence with $T_{\text{spoof}} = 50$ m is deliberate: both thresholds are calibrated to the same noise floor. The range-failure penalty $\alpha_{\text{range}} = 0.10$ is set as the midpoint $(\alpha_{\text{pos}} + \alpha_{\text{neg}})/2$, treating a UWB consistency failure as more severe than a single missed verification (α_{pos}) but less severe than a full failed verification (α_{neg}), since range failures may arise from temporary link degradation rather than deliberate Byzantine behaviour.

E. Dynamic Anchor Selection and Position Relay

TRN-capable UAVs compete for anchor status through the score

$$A_i = w_{\text{rep}} R_i + w_{\text{conf}} C_i + w_{\text{sat}} S_i + w_{\text{fol}} F_i, \quad (10)$$

with default weights (0.4, 0.3, 0.2, 0.1). The weight ordering $w_{\text{rep}} > w_{\text{conf}} > w_{\text{sat}} > w_{\text{fol}}$ reflects a safety-first hierarchy: trustworthiness must dominate performance, which in turn dominates location-dependent factors (imagery quality varies with where the UAV currently is, not what it is), with load balancing as the lowest-priority factor since over-loading is a soft degradation rather than a safety violation. Placing $w_{\text{rep}} = 0.4$ as the dominant term ensures that a compromised peer with degraded reputation $R_i \rightarrow 0$ contributes at most 0.6 to its anchor score even with maximal C_i, S_i, F_i , leaving it below honest candidates whose $R_i \geq T_{\text{quorum}} = 0.6$ adds at least 0.24 from the reputation term alone—an inherent score gap that makes Byzantine-anchor promotion infeasible without first restoring reputation through honest verifications. Given the strict ordering and the unit-sum constraint $\sum w = 1$, the equal-spacing choice (0.4, 0.3, 0.2, 0.1) is adopted as a working default: the strict weight ordering $w_{\text{rep}} > w_{\text{conf}} > w_{\text{sat}} > w_{\text{fol}}$ reflects the safety-first hierarchy, the unit-sum constraint $\sum w = 1$ is met by construction, and the choice is deferred to the Track 2 ablation sweep. The spacing $\Delta w = 0.1$ also exceeds the per-event reputation step $\alpha_{\text{pos}} = 0.05$, ensuring that single-verification reputation jitter cannot swap adjacent priority terms in the score ordering. Sensitivity to the weight vector is part of the Track 2 ablation sweep over $w_{\text{rep}} \in [0.3, 0.5]$, $w_{\text{conf}} \in [0.2, 0.35]$, $w_{\text{sat}} \in [0.15, 0.30]$, $w_{\text{fol}} \in [0.05, 0.20]$ subject to the unit-sum constraint. Here C_i is the TRN confidence (RANSAC inlier ratio), S_i is the imagery coverage quality for the terrain below v_i , and F_i is an inverse load factor. Every $T_{\text{election}} = 60$ s, the M vehicles with the highest A_i are confirmed as active anchors; a standby replaces an active anchor when its score exceeds the incumbent's by a hysteresis $H = 0.1$. The election period $T_{\text{election}} = 60$ s is set to exceed the verification revisit cycle ($\nu_{\text{verify}}^{-1} \approx 45$ s) by 33%, allowing at least one full verification round to complete between elections; this prevents frequent re-elections from disrupting ongoing verifications while remaining responsive to anchor degradation. The hysteresis band $H = 0.1$ corresponds to two consecutive $\alpha_{\text{pos}} = 0.05$ upward reputation steps,

requiring the score gap to represent at least two successful-verification equivalents before triggering a handover. Handover proceeds in three steps (*PROMOTION*, *DEMOTION*, linear-blend) over $T_{\text{handover}} = 10$ s. $T_{\text{handover}} = 10$ s matches the failover total budget of Section IV-F (F1–F5 phases sum to ≈ 8 s nominal, 10 s with 5% packet loss): all anchor-change operations—whether triggered by destruction (failover) or by score-based replacement (planned handover)—share the same architectural timescale, which simplifies the operator’s deployment-time reasoning about transition windows. The lower bound on T_{handover} is set by the cumulative protocol latency of the three-step *PROMOTION* $\rightarrow$ *DEMOTION* $\rightarrow$ linear-blend sequence (UWB neighbour discovery ~ 1 s, role-change acknowledgement round-trip ~ 1 – 2 s, follower-side covariance update propagation through the iSAM2 factor graph ~ 1 s), so T_{handover} must comfortably exceed ~ 4 s to ensure all followers complete their re-association within the blend window. The upper bound is set by the verification revisit cycle: T_{handover} must remain well below $\nu_{\text{verify}}^{-1} \approx 45$ s so that handovers are decisive on a timescale shorter than reputation reassessment. The value 10 s sits an order of magnitude below verification cycle and comfortably above protocol latencies. During the blend, both incumbent and standby publish Distilled State updates with time-weighted weights summing to unity; a longer blend would extend this dual-publishing window, increasing UWB neighbour bandwidth consumption and creating more state-merge events at follower-side factor graphs, while a shorter blend risks discontinuities in follower position estimates between the demoted-incumbent’s last publication and the promoted-standby’s first authoritative state. Sybil protection requires $T_{\text{warmup}} = 6$ successful verifications before anchor eligibility; at $\nu_{\text{verify}}^{-1} \approx 45$ s per cycle, this corresponds to $6 \times 45 = 270$ s (≈ 4.5 min) of observed honest behaviour, exceeding four full election periods, so a freshly-joining Byzantine peer cannot accumulate quorum influence within a single election cycle. These are working defaults.

Followers compute their position as the reputation-weighted average

$$\mathbf{p}_{\text{fol}} = \frac{\sum_j w_j \mathbf{p}_j}{\sum_j w_j}, \quad w_j = \frac{R_j}{\sigma_{\text{hop},j}^2}, \quad (11)$$

where for a k -hop relay chain,

$$\sigma_k = \sigma_{\text{anchor}} (1 + \beta)^k, \quad (12)$$

with $\beta = 0.15$ per hop (per-hop noise growth factor, a conservative working default; see Section III discussion) and σ_{anchor} given by (9). The multiplicative form $\sigma_k = \sigma_{\text{anchor}} (1 + \beta)^k$ models two physical effects that compound at each relay hop in a UWB-trilaterated position chain: (i) additive UWB ranging-measurement variance σ_{range}^2 from the new measurement to the previous hop, and (ii) propagated upstream position uncertainty from the previous hop’s estimate, weighted by a Geometric Dilution of Precision (GDOP) factor that depends on the local anchor configuration. Under perfectly independent ranging and ideal anchor geometry, error propagation is sublinear (*additive variance model*): $\sigma_k = \sqrt{\sigma_{\text{anchor}}^2 + k\sigma_{\text{range}}^2}$,

TABLE II
CONSOLIDATED SYSTEM HYPERPARAMETERS. DERIVATION COLUMN INDICATES WHETHER THE VALUE IS LITERATURE-DERIVED (L), OPERATIONALLY DISCRETIONARY (O), OR A WORKING DEFAULT PENDING SENSITIVITY ANALYSIS (W).

Parameter	Value	Symbol	Derivation
Feature descriptors per PoO	50	N	W
Mode-A compact set size	20	K	W
Signature penalty	0.30	α_{sig}	W
Verification radius	500 m	r_{verify}	W
L2 match threshold	0.7	τ_{match}	L (SuperPoint)
VERIFIED score threshold	0.3	T_{verify}	W
Initial reputation	0.5	R_{init}	W
Positive reputation step	0.05	α_{pos}	W
Negative reputation step	0.15	α_{neg}	W (3:1 ratio)
Temporal decay constant	10^{-3} s^{-1}	λ	W
Staleness cutoff	3600 s	$T_{\text{max_verify}}$	O
Base prior noise	50 m	σ_{base}	W
Noise floor	0.01	ε	W
UWB consistency threshold	50 m	δ_{consist}	W
Range-failure penalty	0.10	α_{range}	W
Anchor-score weights	(0.4,0.3,0.2,0.1)	$w_{1..4}$	W
Election period	60 s	T_{election}	O
Standby-replace hysteresis	0.10	H	W
Handover blend duration	10 s	T_{handover}	O
Sybil warmup count	6	T_{warmup}	W
Per-hop noise growth	0.15	β	W
Max relay hops (ISR)	3	K_{max}	O
Max relay hops (patrol)	5	K_{max}	O
Reject threshold	0.20	T_{reject}	W
Quorum reputation floor	0.60	T_{quorum}	W
Load-balance period	30 s	T_{balance}	O
Load-balance low threshold	30%	P_{low}	O
Load-balance high threshold	150%	P_{high}	O
Spoof position threshold	50 m	T_{spoo}	W
Consecutive spoof count	3	$N_{\text{consecutive}}$	W
Mass-spoofing trigger	30%	P_{mass}	O

growing as $\sqrt{k}$ for large k —this is the theoretical lower bound for independent Gaussian ranging noise. The multiplicative form is not a physical prediction but a *conservative engineering bound*: it dominates the additive model for all $k \geq 1$ when $\beta > 0$, as $(1+\beta)^k > \sqrt{1 + k(\sigma_{\text{range}}/\sigma_{\text{anchor}})^2}$ for typical UWB parameters. Under correlated upstream uncertainty or poor geometry (anchors nearly collinear with the follower), GDOP amplifies propagated variance beyond the sublinear baseline, with the worst case approaching exponential $(1+\beta)^k$ growth. The multiplicative form is therefore the *conservative engineering bound* for the planned non-collinear but topologically constrained swarm. The value $\beta = 0.15$ is adopted as a conservative working default for the per-hop standard-deviation growth factor; it represents a 15% std-dev increase per hop relative to the prior hop, chosen as a round figure within the typical range expected for UWB cross-trilateration under benign (line-of-sight) propagation and pending empirical refit against a measured UWB NLOS dataset in Track 2. From this β , the $K_{\text{max}} = 3$ ISR ceiling follows as a derived quantity (at the reference operating point $\sigma_{\text{anchor}} \approx 58$ m, $R_{\text{anchor}} = 0.85$, $\sigma_3 \approx 88$ m, giving single-anchor $\text{CEP}_{50} = 1.177\sigma_3 \approx 104$ m, slightly exceeding the 100-m NATO-grade target by $\approx 4\%$; under multi-anchor reputation-weighted fusion ($m=2$ – 3) the bound improves to $\text{CEP}_{50} \in [60, 74]$ m, comfortably within the

target); $K_{\max} = 3$ is not used to motivate the choice of β . A sensitivity sweep over $\beta \in \{0.05, 0.10, 0.15, 0.20\}$ is planned in Track 2 and will shift the derived $K_{\max}$ accordingly. The functional form and value of β are subject to empirical refit in Track 2: if measured σ_k versus k fits a sublinear $\sqrt{k}$ form or a different multiplicative rate β' better than the conservative $(1 + 0.15)^k$ assumption, the bound in Proposition 3 relaxes accordingly without affecting the analytical structure. The base noise σ_{base} is a configurable system parameter; we adopt $\sigma_{\text{base}} = 50$ m for consistency across the experiments reported below, with other values applicable to swarm configurations that assume different baseline anchor accuracy. For a well-performing anchor with $R_{\text{anchor}} = 0.85$, $\sigma_{\text{anchor}} = 50/0.86 \approx 58.1$ m; propagation yields $\sigma_1 \approx 66.8$, $\sigma_2 \approx 76.9$, $\sigma_3 \approx 88.4$, and $\sigma_5 \approx 116.9$ m. The 5-hop value exceeds the 100-m target both as σ_5 and as the corresponding $\text{CEP}_{50} \approx 138$ m; for NATO-grade ISR we therefore recommend $K_{\max} = 3$ under the default multi-anchor fusion regime, reserving $K_{\max} = 5$ for lower-precision patrol missions. The patrol value $K_{\max} = 5$ is reported as an upper operational envelope, not a precision recommendation: at $R_{\text{anchor}} = 0.85$ the per-hop derivation yields $\sigma_5 \approx 117$ m ($\text{CEP}_{50} \approx 138$ m), exceeding the 100 m NATO-grade ISR target. The intermediate $K_{\max} = 4$ is not separately reported because it is not operationally distinct from $K_{\max} = 5$ for precision purposes; $K_{\max} = 3$ under multi-anchor fusion is the maximum strictly ISR-compliant depth (single-anchor at $k = 3$ exceeds the CEP target by $\approx 4\%$ and is flagged for deployment-time review). The patrol use case accepts precision on the order of ~ 150 m CEP or coarser in exchange for a $1.67\times$ greater effective swarm reach (relay radius 2.5 km at $K_{\max} = 5$ versus 1.5 km at $K_{\max} = 3$, given the 500-m UWB hop range). For anchors operating below the reference $R_{\text{anchor}} = 0.85$, σ_k scales as $1/(R_{\text{anchor}} + \varepsilon)$ per (9), so both bounds tighten proportionally; the precision tolerance is a per-mission operator decision and the relay-depth ceiling is set deployment-time as a function of the mission's precision class. If R_{anchor} drops from 0.85 to 0.30, σ_{anchor} rises to ~ 161 m and all downstream followers automatically receive degraded weights through (11) without any explicit control signal: *reputation flows down the hierarchy through noise covariance*.

F. Automatic Failover Protocol

Anchor loss (destruction, jamming, or reputation falling below T_{reject}) triggers a five-phase protocol executing within 10 s:

- **F1 Detection** (0–5 s): a watchdog declares anchor unavailability after $T_{\text{timeout}} = 5$ s without a DistilledState message. During the timeout, orphaned followers propagate by IMU dead reckoning, accumulating <5 m drift at 30 m/s.
- **F2 REASSIGN_REQUEST** (5–6 s): each orphaned follower broadcasts its device ID, last known position, velocity, battery state, and sensor capabilities.

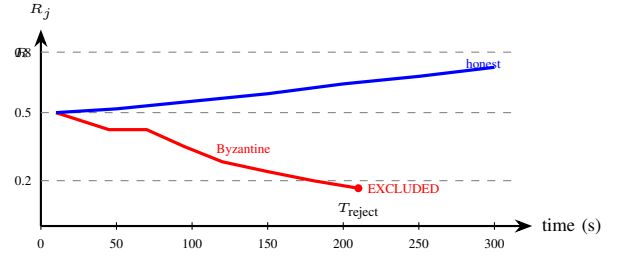


Fig. 2. Schematic of the analytically predicted reputation trajectory under a naïve Byzantine attack (Section VI-C; empirical validation deferred). The compromised UAV's reputation decays after each failed Proof-of-Observation verification while honest peers continue to accumulate trust. Permanent exclusion is predicted within the worst-case upper bound $T_{\text{exclusion}} \lesssim 215$ s (Proposition 3); the schematic marks $T \approx 210$ s, at which three independent honest verifiers have each driven R_j below $T_{\text{reject}} = 0.2$. The per-step drop $R_4 : 0.5 \rightarrow 0.425$ illustrated at $T=45$ s corresponds to an illustrative verifier weight $R_k = 0.5$ (so that $\alpha_{\text{neg}} \cdot R_k = 0.075$); the operational deployment uses R_k drawn from the honest reputation distribution, and the Proposition 3 bound uses the conservative substitution $R_k \rightarrow \bar{R}_{\text{honest}} = 0.53$ for the analytical estimate.

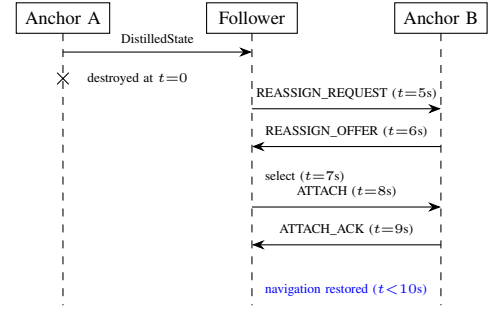


Fig. 3. Failover sequence upon anchor destruction. Dead-reckoning drift during the 5-second timeout is <5 m at 30 m/s cruise speed.

- **F3 REASSIGN_OFFER** (6–7 s): surviving anchors respond with their ID, position, reputation, and available capacity.
- **F4 Selection** (7–8 s): the follower computes
$$S_{\text{reassign}} = \alpha_{\text{prox}}/d + \alpha_{\text{rep}}R + \alpha_{\text{cap}}(1 - L), \quad (13)$$
with defaults (0.4, 0.35, 0.25).
- **F5 Attachment** (8–10 s): ATTACH/ATTACH_ACK exchanges position and covariance.

Byzantine rejection proper removes all factors of UAV j from the local graph when $R_j < T_{\text{reject}} = 0.2$; permanent exclusion additionally requires M independent UAVs, each with $R > T_{\text{quorum}} = 0.6$, to confirm the exclusion. From Proposition 3, convergence is predicted at $T_{\text{ind}} = (R_{\text{init}} - T_{\text{reject}})/(\alpha_{\text{neg}} \bar{R}_{\text{honest}} \nu_{\text{verify}}) \approx 170$ s; with $R_{\text{init}} = 0.5$ the gap $R_{\text{init}} - T_{\text{reject}} = 0.3$ drives the numerator. A smaller T_{reject} would increase convergence time, while a larger T_{reject} would increase false exclusions of honest peers; $T_{\text{reject}} = 0.2$ keeps the honest-peer safety margin above 2 environment-induced failures at $R_k = 0.5$ (see Proposition 1 analysis). The quorum reputation floor $T_{\text{quorum}} = 0.6$ requires a verifier to have earned at least two successful verifications above the initial state ($T_{\text{quorum}} = R_{\text{init}} + 2\alpha_{\text{pos}} = 0.5 + 0.10 = 0.6$), so

a freshly-joined peer cannot self-confirm into the quorum; the minimum defensible value is $T_{\text{quorum}} > R_{\text{init}} = 0.5$, and $T_{\text{quorum}} = 0.6$ provides a minimal earned-trust barrier. Coincidentally $T_{\text{quorum}} = 3 \times T_{\text{reject}}$, but the binding constraint is the margin above R_{init} , not the factor of 3; the factor of 3 is a working default providing a clear separation between verifier quality and exclusion threshold. The default $M = 3$ tolerates $f \leq M - 1 = 2$ colluding Byzantine peers irrespective of n ; safety for larger f requires M to scale with f , as analysed in Section V. Fig. 3 summarises the timing.

G. Load Balancing and GNSS Spoofing Detection

Every $T_{\text{balance}} = 30$ s, anchors broadcast a STATUS field (appended to the standard Distilled State message of Section IV-D) including their follower count. The load-balancing period is set to one-half of the election period ($T_{\text{election}} = 60$ s), allowing two load-balance checks per election cycle; at $\nu_{\text{verify}}^{-1} \approx 45$ s, a 30 s check interval falls between consecutive verifications. The target is $T_{\text{target}} = N_{\text{total}}/N_{\text{anchors}}$. If an anchor's count is below $P_{\text{low}}T_{\text{target}}$ (30%) or above $P_{\text{high}}T_{\text{target}}$ (150%), a redistribution protocol (TRANSFER_OFFER $\rightarrow$ TRANSFER_ACCEPT $\rightarrow$ REASIGN_COMMAND $\rightarrow$ smooth linear-blend handover over $T_{\text{handover}} = 10$ s) transfers followers from overloaded to underloaded anchors. The asymmetric band reflects the asymmetric cost of under- vs over-loading: an overloaded anchor ($> 150\%$ of target) risks queue overflow and message drops, while an underloaded anchor ($< 30\%$ of target) wastes TRN capacity without causing message loss. Overload headroom: $P_{\text{high}} - 1.0 = 0.50$; underload floor: $1.0 - P_{\text{low}} = 0.70$. These are working defaults consistent with general load-balancing practice [59].

GNSS-spoofing detection operates at 1 Hz on each UAV that still has a GNSS receiver (relevant when GNSS is only partially denied or during the transition into full-denial operation). The vehicle compares its GNSS-reported position with a UWB-trilaterated position from at least three neighbours with $R > T_{\text{min}}$, where $T_{\text{min}} = T_{\text{reject}} = 0.20$ excludes peers that are already excluded by the reputation system from participating in the consistency cross-check. If $\|\mathbf{p}_{\text{GNSS}} - \mathbf{p}_{\text{UWB}}\| > T_{\text{spooft}} = 50$ m for $N_{\text{consecutive}} = 3$ successive measurements, GNSS is marked unreliable and the affected vehicle falls back to reputation-weighted UWB position. The spoofing detection threshold is set equal to $\sigma_{\text{base}} = 50$ m so that UWB-measured positions deviating from the claimed GNSS position by more than one prior standard deviation are flagged as anomalous; UWB ranging error at 500 m range is typically 10–30 cm ($1\text{-}\sigma$), so $T_{\text{spooft}} = 50$ m corresponds to $\sim 170\text{--}500 \sigma_{\text{UWB}}$, far above any legitimate ranging uncertainty. $N_{\text{consecutive}} = 3$ requires three consecutive exceedances, reducing false-positive rate to approximately p^3 where p is the per-sample false-alarm rate; three is the minimum count providing a causal chain (removes single-sample noise artefacts) while remaining below the $T_{\text{warmup}} = 6$ threshold (so an attacker cannot “pass” warmup before being flagged). When the proportion of spoofed UAVs exceeds $P_{\text{mass}} = 30\%$, the entire swarm transitions to a *full*

GNSS-denial mode: GNSS is disabled globally and the swarm relies exclusively on TRN anchors with UWB relay. This mass-spoofing threshold is set well above the per-Proposition-1 maximum Byzantine fraction ($f/n \leq 2/200 = 1\%$ under default $M = 3, f = 2$): a coordinated attack on 30% of the swarm constitutes a scenario outside the individual-exclusion threat model, where full-GNSS-denial mode is the appropriate response. The 30% figure is chosen just below the classical Byzantine bound $f < 1/3 \approx 33\%$: when $> 30\%$ of the swarm reports spoofed positions, the collective deviation exceeds the Byzantine-fault-tolerant range of the reputation mechanism (which tolerates $f < M - 1$ individual colluders, not 30% of 200 UAVs simultaneously), so global GNSS-denial mode is the appropriate response. This is a working default, operator-configurable. By design, GNSS spoofing does not affect navigation reputation R_i ; only failed Proof-of-Observation verifications decrement reputation, so a UAV spoofed by an external emitter is not punished for an attack it did not launch.

V. THEORETICAL ANALYSIS

A. Quorum Safety

Proposition 1 (Quorum safety against naïve Byzantine peers): Let M be the permanent-exclusion quorum size. Assume (i) naïve Byzantine peers fail every Proof-of-Observation verification (they claim positions they have not physically observed); (ii) honest peers correctly issue VERIFIED or UNVERIFIED verdicts; (iii) each UAV maintains its reputation state R_j locally from its own verifications, without gossip-based propagation of reputation values between peers; and (iv) the per-peer rate of environment-induced UNVERIFIED verdicts on honest targets, ρ_{env} (events per second), satisfies $\rho_{\text{env}} < \alpha_{\text{pos}} \nu_{\text{verify}} / \alpha_{\text{neg}}$, so that honest peers accumulate VERIFIED events faster than environmental penalties drive R toward T_{reject} . Condition (iv) is a *rate condition on the environmental failure regime*; the Remark below quantifies what happens when it is violated (e.g., persistent cloud cover or night-time operation). The proposition provides no safety guarantee against environment-induced false exclusion when (iv) is violated, only against deliberate Byzantine attack; and (v) $R_{\text{init}} < T_{\text{quorum}}$, so that a freshly-joined Byzantine peer cannot be self-confirmed into the quorum by virtue of its initial reputation alone (the default values $R_{\text{init}} = 0.5 < T_{\text{quorum}} = 0.6$ satisfy this). If $f \leq M - 1$, no set of colluding Byzantine peers can trigger false permanent exclusion of an honest UAV.

Sketch: Permanent exclusion requires M independent UAVs, each with $R > T_{\text{quorum}}$, to declare the target $R_j < T_{\text{reject}}$. Naïve Byzantine peers cannot raise their reputation above T_{quorum} because every verification attempt against them yields UNVERIFIED (assumption (i)); each UAV tracks this locally from its own verifications without reputation gossip (assumption (iii)), so collusion to inflate a Byzantine peer's remote reputation is blocked; and the initial reputation $R_{\text{init}} < T_{\text{quorum}}$ (assumption (v)) ensures no freshly-joined peer enters the quorum immediately. Therefore any false quorum contains at most f Byzantine members; if $f < M$, at least one honest member would need to declare a false UNVERIFIED

on an honest target, which contradicts assumption (ii). Assumption (iv) further ensures that an honest target's reputation does not drift below T_{reject} from environmental causes alone, eliminating the alternative failure mode of false exclusion driven by accumulated environment-induced UNVERIFIED events. ■

Remark 1: If Assumption (iv) is violated (e.g., dense cloud cover or night-time operation rendering terrain features undetectable so ρ_{env} rises sharply), Proposition 1 still holds against deliberate Byzantine attack but does not prevent environment-induced false exclusion of an otherwise-honest UAV. In that operational regime, operator intervention or automatic mode switch to INS-only localisation is required.

Proposition 2 (Parameter scaling): Assume $n \geq M + f + 1$. Then the default configuration $M = 3$ tolerates at most $f = 2$ colluding Byzantine peers. The precondition $n \geq M + f + 1$ ensures that M honest verifiers can always be assembled independently of the f Byzantine nodes, with one additional honest peer as a reserve against transient communication failure; the formal mathematical minimum is $n \geq M + f$ (the quorum can be exactly filled by honest peers with none to spare), but at this boundary any single transient unavailability blocks quorum assembly, so $n \geq M + f + 1$ is the recommended operational floor. For $M = 3$, $f = 2$ this gives $n \geq 6$. At $n = 5$, only $5 - f = 3$ honest peers remain, which exactly meets $M = 3$ but leaves no spare verifier; any additional failure violates the constraint. The precondition $n \geq M + f + 1$ must be enforced by the deployment operator. For swarms that require tolerance of more adversaries, M must be configured as $M \geq f + 1$, subject to the further constraint that at least M honest peers must accumulate $R > T_{\text{quorum}}$ before exclusion becomes possible.

The classical Byzantine impossibility bound $f < \lfloor (n - 1)/3 \rfloor$ [12] (with related graph-theoretic robustness conditions developed by [24] for the W-MSR setting) is *achievable* in our system—but only if M is scaled accordingly; it is not automatic from swarm size. This is a design trade-off: a small M gives fast exclusion in small swarms, a large M gives stronger guarantees in large swarms at the cost of slower exclusion times.

Strategic adversaries. Proposition 1 assumes naïve Byzantine behaviour. Strategic Byzantine adversaries—which the trust-based multi-agent literature has long recognised as difficult [20], [22]—can in principle maintain $R > T_{\text{quorum}}$ between observable attacks. Our system provides partial mitigation through (i) the UWB consistency check (Section IV-D), which catches range–position inconsistencies independent of PoO; (ii) the asymmetric reputation rule, which amplifies any single failed verification; and (iii) the temporal-decay mechanism, which prevents historical trust from indefinitely shielding an adversary. A formal treatment of strategic Byzantine resilience is beyond the scope of this paper and is left to future work.

B. Time to Naïve Byzantine Exclusion

Proposition 3 (Convergence bound): Assume (i) $R_j = R_{\text{init}}$ at the moment the target j begins Byzantine behaviour (i.e.

the bound applies at the moment of compromise, before any honest-trust accumulation; if j has accumulated $R_j > R_{\text{init}}$ during a warm-up window of duration T_{warmup} , the bound increases proportionally to $(R_j - T_{\text{reject}})/(R_{\text{init}} - T_{\text{reject}})$, replacing the numerator of Eq. (14)); and (ii) the per-event verifier reputation weight is approximated by its honest-equilibrium value $R_k \approx \bar{R}_{\text{honest}}$ (the substitution $R_k \leftarrow \bar{R}_{\text{honest}}$ in Eqs. (1)–(3) collapses the per-event modulation to a constant drift coefficient). Let ν_{verify} denote the mean rate (in s^{-1}) at which honest UAVs fly over regions claimed by a Byzantine peer, and let $\bar{R}_{\text{honest}}$ be the average honest reputation. (The verification radius $r_{\text{verify}} = 500$ m of Section IV-B is a distinct quantity.) For a naïve Byzantine peer, the expected time at a single honest verifier to drive R_j below T_{reject} is

$$T_{\text{ind}} \approx \frac{R_{\text{init}} - T_{\text{reject}}}{\alpha_{\text{neg}} \bar{R}_{\text{honest}} \nu_{\text{verify}}}. \quad (14)$$

Permanent exclusion additionally requires M independent honest verifiers to reach the threshold. Because the verifier set is not strictly independent—different verifiers may overlap on the same claimed region and start their verification sequences at correlated times—the additional quorum delay is bounded below by $(\rho - 1)/\nu_{\text{verify}}$, where $\rho \geq 1$ captures inter-verifier overlap. A closed-form expression for ρ in realistic topologies is beyond the scope of this paper; empirical characterisation of the quorum delay is the purpose of Experiment 3 of the proposed evaluation methodology (Section VI).

Substituting representative parameter choices $R_{\text{init}} = 0.5$, $T_{\text{reject}} = 0.2$, $\alpha_{\text{neg}} = 0.15$, and the design assumptions $\bar{R}_{\text{honest}} = 0.53 \approx R_{\text{init}} + 0.5\alpha_{\text{pos}} = 0.5 + 0.025 = 0.525$ (one half-step above the initial state; a conservative lower estimate bounded between $R_{\text{init}} = 0.5$ and ≈ 0.6 ; the actual equilibrium via the temporal-decay ODE is deferred to Track 2; a higher equilibrium would yield faster exclusion) and $\nu_{\text{verify}}^{-1} \approx 45$ s (design input; corridor-geometry lower bound: anchor traversal of one coverage cell diameter $2r_{\text{verify}} = 1000$ m at $v_p = 30$ m/s takes $1000/30 \approx 33$ s; the additional ≈ 12 s is a conservative overhead budget for UWB handshake round-trip ($\lesssim 2$ s), TRN fix latency ($\lesssim 5$ s at 0.1 Hz fix rate), and scheduling jitter ($\lesssim 5$ s); exact characterisation deferred to Track 2) give $T_{\text{ind}} \approx 170$ s. A closed-form geometric derivation of ν_{verify}^{-1} depends strongly on the specific anchor routing pattern; a tractable analytical model is left to future work. Adding the inter-verifier quorum delay $(\rho - 1)/\nu_{\text{verify}}$ under the working assumption $\rho \leq 2$ (i.e. inter-verifier overlap adds at most one additional verification interval at the $M = 3$ default, consistent with the patrol-topology revisit cycle of Section VI-A) and noting that T_{ind} is reported in seconds via ν_{verify} in s^{-1} (whereas the simulator's per-cell T_e is reported in rounds, with one round corresponding to one observation step in the abstract simulator), yields an analytical *estimated worst-case upper bound* $T_{\text{exclusion}} \lesssim 215$ s under these conservative parameter substitutions; the actual exclusion time is expected to be lower whenever the operational equilibrium $\bar{R}_{\text{honest}} > 0.53$ holds (e.g., $\bar{R}_{\text{honest}} \rightarrow 1$ in the high-trust limit gives $T_{\text{ind}} \approx 90$ s and $T_{\text{exclusion}} \lesssim 135$ s). Empirical characterisation of the actual

exclusion-time distribution relative to this worst-case bound is the purpose of Experiment 3 of the proposed evaluation methodology (Section VI-C).

C. Multi-Hop Error Propagation

From (12), the k -hop standard deviation satisfies $\sigma_k = \sigma_{\text{anchor}}(1 + \beta)^k$. The mean-squared error of the fused position (11) at a follower receiving m anchor contributions with weights w_j is

$$\text{MSE} = \frac{\sum_j w_j^2 \sigma_j^2}{\left(\sum_j w_j\right)^2}, \quad (15)$$

which follows directly from the standard weighted-average variance identity for independent Gaussian observations [55]. The independence assumption is only approximate in the deployed system: multiple anchor position estimates are jointly maintained inside a shared iSAM2 Bayes-tree factor graph and share UWB-ranging factors, so their residual covariances are partially correlated; the MSE expression above is therefore an *optimistic* (lower-bound) estimate of the achievable CEP under reputation-weighted fusion. The empirical CEP measured in Experiment 1 of Section VI will be the binding deployment number; the analytical bound is reported here for design-time parameter selection. For $\sigma_{\text{anchor}} = 58.1$ m (i.e., $R = 0.85$, $\sigma_{\text{base}} = 50$ m) and $\beta = 0.15$, the corresponding $\text{CEP}_{50} = 1.177\sigma_k$ for the single-anchor case crosses the 100-m NATO threshold between $k = 2$ ($\text{CEP} \approx 90$ m) and $k = 3$ ($\text{CEP} \approx 104$ m, $\approx 4\%$ over); under default multi-anchor fusion ($m=2-3$), the bound at $k = 3$ improves to $\text{CEP} \in [60, 74]$ m, well within the target. Our recommendation $K_{\text{max}}=3$ is therefore predicated on the multi-anchor fusion regime that is the system's default operating mode; single-anchor relay at $k = 3$ is marginally non-compliant and should be flagged at deployment time.

D. Communication Complexity

Each UAV broadcasts a Distilled State message at 1 Hz, totalling approximately 185 bytes. The payload composition is: position $\langle \text{lat}, \text{lon}, \text{alt} \rangle$ as three doubles (24 B), velocity $\langle v_x, v_y, v_z \rangle$ as three floats (12 B), 3×3 symmetric covariance summary (6 floats = 24 B), timestamp (8 B), UAV identifier (8 B), compact reputation/state-flag block (20 B), Ed25519 signature (64 B), and framing overhead (type byte, length, sequence number, CRC, padding ≈ 25 B), summing to ≈ 185 B. For a typical UWB neighbourhood of $n_{\text{local}} \approx 10$ UAVs (within ~ 100 m range, not the full $n = 200$ swarm), each vehicle receives $n_{\text{local}} - 1 = 9$ inbound messages per second, giving an inbound bandwidth of $9 \times 185 \text{ B} \times 8 \text{ bit/B} \times 1 \text{ Hz} \approx 13.3 \text{ kbit/s}$. A Signed Observation (~ 180 – 200 B; see Section IV-B) is transmitted only on TRN fixes at 0.1 Hz, adding negligible overhead. No consensus round-trip is required in the critical navigation path, distinguishing our approach from blockchain-based schemes whose route-commitment finality is on the order of seconds per round, a property of the

blockchain settlement layer [13], [14]. Signal-processing approaches for single-UAV spoofing detection (e.g., deep CNN-BiLSTM frameworks [66]) do not address cooperative swarm-wide verification and therefore do not replace the cross-peer architecture proposed here.

VI. PROPOSED EVALUATION METHODOLOGY

This section specifies the simulation methodology designed to validate the analytical predictions of Section V. The methodology has not yet been executed: empirical results are deferred to future work in a Gazebo/ROS2-based framework currently under development. Each experiment below states (i) the configuration, (ii) the metric of interest, (iii) the analytical prediction (where the theory provides one), and (iv) the pass criterion. Reporting the planned methodology in this paper rather than ad-hoc post-hoc results is intentional: it pre-registers what the proposed system would measure and how a fair comparison against prior art could be conducted.

A. Planned Evaluation Setup

The intended scenario is a swarm of $n = 200$ UAVs over a 200×200 km rural corridor at 300 m altitude and 30 m/s cruise speed: five UAVs TRN-capable (active anchors), two TRN-capable standby, and the remaining 193 followers. Coverage is achieved via a mobile-anchor patrol model rather than static deployment [60]: each anchor sweeps the corridor, providing follower updates at a revisit rate of $\nu_{\text{verify}}^{-1} \approx 45$ s. The operative coverage metric is the maximum revisit interval experienced by any follower, not instantaneous geometric overlap; at any instant, 5 anchors at 500 m UWB range cover $5\pi(0.5)^2 \approx 4 \text{ km}^2$ of the $40\,000 \text{ km}^2$ corridor ($\approx 0.01\%$), which is sufficient because followers receive periodic updates during anchor sweeps rather than requiring continuous line-of-sight coverage. The anchor count of five is a design input: with $K_{\text{max}} = 3$ relay hops and a per-hop range of 500 m, the maximum relay radius is $3 \times 500 = 1500$ m, so a single anchor can serve followers within this radius at any instant; two standby anchors provide redundancy against single-point anchor failure as required by Experiment 2. At any instant, $K_{\text{max}} = 3$ relay hops give a single-anchor coverage footprint of $\pi(1.5)^2 \approx 7 \text{ km}^2$ and five anchors a combined $\approx 35 \text{ km}^2$, i.e. $\approx 0.09\%$ of the $40,000 \text{ km}^2$ corridor; full corridor coverage therefore requires that the union of the five anchors' patrol routes intersects every follower's position within the revisit cycle $\nu_{\text{verify}}^{-1} \approx 45$ s. The specific patrol-route geometry that achieves this (e.g., evenly-spaced longitudinal sweeps, lawnmower partitions, or an outer-loop / inner-loop split between active and standby anchors) is a deployment-time decision treated as out of scope here; the analytical scenario assumes the followers are distributed such that the above revisit condition is met by construction, and Track 2 will validate this against a concrete patrol-route schedule. GNSS is fully denied in the default setup (the spoofing experiment uses a partial-GNSS configuration, see Experiment 4). Planned terrain imagery is Sentinel-2 at 10 m GSD (bands B2, B3, B4); Sentinel-2 is the standard open-access satellite imagery source

for large-area TRN research [58] and its 10 m multispectral bands (290 km swath, 5-day revisit at high latitudes) provide sufficient resolution for the SuperPoint feature-extraction pipeline at the corridor scale. Higher-resolution commercial sources (e.g., 0.5 m imagery) are not used because their licensing restrictions and per-tile cost are incompatible with the open research framing of this paper. Sensor noise parameters of the planned simulator are: MEMS-grade IMU ($10^\circ/\text{h}$ drift), UWB ± 0.1 m, TRN CEP ~ 35 m, barometric altitude ± 2 m (MEMS barometric altimeter working default; MEMS barometers typically specify $\pm 1\text{--}3$ m absolute accuracy at room temperature; Track 2 hardware selection will pin the specific sensor datasheet). Planned mission duration: 2 hours per run. The intended implementation is a Python/C++ simulator with iSAM2 (GTSAM 4.2) [52]. iSAM2 is chosen over batch solvers (g2o, Ceres) and sliding-window SAM because its Bayes-tree incremental update mechanism provides amortised $O(\log n)$ complexity per new measurement by updating only the affected portion of the factor graph; batch solvers would require a full re-linearisation sweep on each update, which does not scale to 200 nodes at 45-second intervals. Each experiment is to be repeated with 25 Monte Carlo seeds (seed values $s_i = i$ for $i \in \{1, 2, \dots, 25\}$), using the NumPy default random generator with `numpy.random.default_rng(s_i)` for statistical characterisation; the seed count is chosen to give a standard error of the mean within roughly 5% of typical metric magnitudes under the assumption of seed-independent Gaussian noise, and will be adjusted upward in the actual run if observed variance is larger.

The planned simulation stack (Python orchestration layer interfacing a C++/GTSAM 4.2 back-end) is expected to require on the order of 2 hours wall-clock time per seed on a modern workstation-class CPU (e.g., 8-core, 3.5 GHz), yielding approximately $25 \times 6 \approx 150$ core-hours for the full evaluation suite. These estimates are provisional and will be refined once the Track 2 simulation framework is implemented; the Section V pass criteria do not depend on achieving any particular runtime.

SwarmRaft baseline (planned comparison). The intended comparison is against an author reimplement of SwarmRaft [2] configured with author-chosen parameters (5-peer leader group, 10-s leader re-election period, median-based position correction); these parameters are not specified in the original SwarmRaft paper. The values are working defaults selected to match the conservative end of the Raft consensus parameter space [1]: Raft’s original evaluation used election timeouts of 150–300 ms for LAN deployments; we scale to 10 s to account for the higher latency of long-range UAV RF links and to ensure leader stability across the $\nu_{\text{verify}}^{-1} \approx 45$ s verification cycle. The 5-peer group is the minimum Raft quorum size ($\lfloor 5/2 \rfloor + 1 = 3$ for majority) that provides two-fault tolerance. These parameters are subject to sensitivity analysis in Track 2. The same UWB/TRN sensor-noise model is used for the baseline as for the proposed system. Because SwarmRaft does not maintain persistent reputation across rounds, “time-to-exclusion” for the baseline is to be measured

as the wall-clock time at which the fraction of rounds in which a target peer’s report is overruled by the median exceeds 90% for ten consecutive elections—a functional analogue of exclusion. Other reasonable definitions yield similar ordering. A head-to-head comparison against the original SwarmRaft codebase remains future work.

B. Proposed Experiment 1: CEP vs. Relay Depth

Configuration: nominal honest-peer operation with $\sigma_{\text{base}} = 50$ m; reputation-weighted fusion across $m = 2\text{--}3$ anchors per follower typical of the planned topology (see (11)). The value $\sigma_{\text{base}} = 50$ m is set as a conservative prior slightly exceeding the TRN CEP specification of ~ 35 m: a 50 m Gaussian 1σ corresponds to $\text{CEP} \approx 50 \times 1.177 \approx 59$ m (using the $\text{CEP} \approx 1.177\sigma$ relation for a circular normal distribution), providing a starting prior that the iSAM2 smoother contracts as observations accumulate. Sensitivity to σ_{base} is deferred to Track 2. **Metric:** CEP at each relay level $k \in \{0, 1, 2, 3, 5\}$, min–max across 25 seeds. Level $k = 4$ is omitted as it provides no qualitatively distinct operating regime between $k = 3$ (ISR ceiling) and $k = 5$ (patrol ceiling); the chosen set brackets each K_{max} with one level on each side of the transition. **Analytical prediction:** Table III gives the CEP_{50} ranges derived from (11)–(12) with $\beta = 0.15$, using the standard $\text{CEP}_{50} = 1.177\sigma_k$ relation for a 2D circular Gaussian; lower bounds assume $m = 3$ -anchor reputation-weighted fusion and upper bounds the single-anchor case, with $\sqrt{m}$ scaling between them. Inter-seed variance bounds will be reported alongside the empirical CEP in Track 2. **Pass criterion:** under the default multi-anchor fusion configuration ($m = 2\text{--}3$), levels up to $k=3$ remain under the 100-m NATO threshold across all 25 seeds (Table III predicts ≤ 74 m at $k = 3$ for $m = 2$); $k=5$ exceeds the threshold in the high-percentile tail, confirming the theoretical analysis of Section V-C. Single-anchor relay at $k = 3$ is marginally non-compliant ($\text{CEP} \approx 104$ m, $\approx 4\%$ over) and is reported separately as a worst-case bound rather than the operational target. On this basis the architecture specifies $K_{\text{max}}=3$ for ISR missions and reports $K_{\text{max}}=5$ only as an upper operational envelope. An INS-only dead-reckoning baseline is expected to reach the kilometre scale over a 2-hour mission given the MEMS-grade IMU drift, against which the relay scheme is predicted to yield more than an order-of-magnitude reduction.

C. Proposed Experiment 2: Anchor Failover

Configuration: at $T=15$ min, anchor #3 (bearing approximately 39 followers given the planned topology) is destroyed by simulated kinetic impact. The trigger time $T = 15$ min is chosen to occur (a) well after the system warmup phase ($T_{\text{warmup}} \times \nu_{\text{verify}}^{-1} = 6 \times 45 \text{ s} = 270 \text{ s} \approx 4.5$ min; $T = 15$ min is $\approx 3.3\times$ the warmup duration), and (b) past one temporal-decay half-life ($\ln 2/\lambda \approx 11.5$ min), so anchor #3 has accumulated approximately $(15 - 4.5)/(\nu_{\text{verify}}^{-1}/60) \approx 14$ post-warmup verification opportunities per verifier-peer and operates in the asymptotic regime of both reputation accumulation and decay—representing the loss of an *established* anchor

TABLE III

ANALYTICAL CEP PREDICTION VS. RELAY HOP COUNT, WITH $\sigma_{\text{BASE}}=50$ M, $R_{\text{ANCHOR}} = 0.85$ ($\sigma_{\text{ANCHOR}} \approx 58.1$ M), AND $\beta = 0.15$. BOUNDS ARE COMPUTED VIA THE STANDARD $\text{CEP}_{50} = 1.177\sigma_k$ RELATION FOR A 2D CIRCULAR GAUSSIAN; LOWER BOUNDS CORRESPOND TO $m=3$ -ANCHOR REPUTATION-WEIGHTED FUSION ($\text{CEP}_{50} \approx 1.177\sigma_k/\sqrt{3}$), UPPER BOUNDS TO THE SINGLE-ANCHOR CASE ($m=1$). EMPIRICAL VALIDATION IS THE PURPOSE OF EXPERIMENT 1.

Level	Role	Position Source	Predicted CEP (m)
0	Anchor (TRN)	Cam + imagery + GTSAM	30–50
1	Direct follower	Anchor + UWB	45–79
2	Relay (2 hops)	Relay + UWB	52–90
3	Deep (3 hops)	2nd relay + UWB	60–104
5	Maximum (5 hops)	5th relay + UWB	79–138

rather than a recently-promoted candidate. The remaining $120 - 15 = 105$ min provides a long post-failover observation window for characterising swarm CEP recovery after orphan reassignment. Earlier trigger times ($T < 10$ min) would test recently-promoted anchors with non-asymptotic reputation, conflating failover dynamics with bootstrap dynamics; later trigger times would compress the observation window without adding scenario value. The REASSIGN protocol is expected to redistribute orphans across the remaining anchors. **Metric:** total switching time (anchor destruction to all-followers-reattached) and maximum dead-reckoning drift during the interval, mean $\pm$ std across 25 seeds; same statistics under injected 5% packet loss. The 5% packet-loss injection is a working default chosen to match the Experiment 2 analytical-prediction stress regime: “injecting 5% packet loss is predicted to push the mean toward but below the 10-s design budget” (Section VI-B Analytical prediction). The value 5% therefore tests the failover protocol at the operating point where the design prediction expects a near-budget but in-budget outcome, providing a direct empirical check of the analytical model. The lower edge of the operating envelope is clean line-of-sight UWB (commercial DW3000-class ranging radios report typical loss well below 1% at sub-500-m range under low-mobility conditions); the upper edge is set by the F1–F5 budget itself, beyond which retransmission overhead saturates the 10 s budget and the test transitions from operating-envelope to failure-mode characterisation. A sensitivity sweep over $\{1\%, 5\%, 10\%, 20\%\}$ is planned in Track 2 to characterise the full envelope; the same sweep will calibrate the packet-loss model against measured airborne UWB link statistics, which the available published literature does not report at the per-link granularity required for direct substitution at this stage. **Analytical prediction:** failover phases F1–F5 (Section IV-F) sum to a cumulative latency on the order of 8 s under nominal UWB, with the F1 watchdog interval (≈ 5 s) the dominant term; injecting 5% packet loss is predicted to push the mean toward but below the 10-s design budget. **Pass criterion:** total switching time < 10 s under both nominal and 5%-packet-loss conditions, with no follower left orphan after the protocol completes; the per-phase intervals are recorded individually so the dominant contributor to any over-budget run can be identified. The “no orphan” criterion applies seed-by-seed: in

each of the 25 seeds, every follower attached to the destroyed anchor must successfully reattach to a surviving anchor within the 10-s budget; partial failures (e.g., one orphan in one seed) constitute a failed pass and trigger investigation.

D. Proposed Experiment 3: Byzantine Detection and Exclusion

Configuration: at $T=10$ s, anchor #4 is compromised (naïve Byzantine) and begins transmitting Signed Observations with spoofed positions displaced by 2 km. The 2-km spoofed position displacement is a working default chosen to lie well above all detection thresholds and well below the operational-area scale: (a) well above the UWB consistency threshold $\delta_{\text{consist}} = 50$ m, so the UWB cross-check flags the displacement independently of PoO verification; (b) well above the worst-case per-hop position uncertainty $\sigma_3 \approx 88$ m at the $K_{\text{max}} = 3$ relay bound ($2 \text{ km} \approx 23 \sigma_3$), so the displacement cannot be washed out by a verifier’s own position noise regardless of its hop distance from a trusted reference; (c) well above the TRN CEP of ~ 35 m, ensuring the claimed-terrain region’s descriptors differ substantively from those at the anchor’s actual location—this is the primary PoO detection path; and (d) well below the 200-km corridor scale ($2 \text{ km} = 1\%$ of corridor), so the displaced claimed position remains within the simulated terrain database and within reach of patrolling honest peers who can verify it. Both detection paths (UWB consistency and PoO descriptor mismatch) fire independently at this displacement; values between ~ 200 m and ~ 5 km yield qualitatively identical Byzantine-exclusion behaviour. The specific 2-km value is reported for reproducibility rather than as a derived bound. Sensitivity over $\{100 \text{ m}, 500 \text{ m}, 2 \text{ km}, 10 \text{ km}\}$ is planned in Track 2. Signatures remain valid (key compromised with the UAV), but descriptor hashes do not match the claimed terrain. Note: the compromise trigger at $T = 10$ s precedes the $T_{\text{warmup}} = 6$ successful verifications required for Sybil protection ($6 \times \nu_{\text{verify}}^{-1} \approx 270$ s). This intentionally tests the pre-warmup worst-case: the analytical bound (Proposition 3) applies with $\bar{R}_{\text{honest}}$ estimated from R_{init} , representing maximum attacker advantage at the moment of compromise. **Metric:** time from compromise to permanent exclusion ($R_4 < T_{\text{reject}}$ confirmed by $M=3$ independent verifiers), mean $\pm$ std across 25 seeds. **Analytical prediction:** Section V-B’s worst-case upper-bound estimate $T_{\text{exclusion}} \lesssim 215$ s under the conservative parameters $R_{\text{init}}=0.5$, $T_{\text{reject}}=0.2$, $\alpha_{\text{neg}}=0.15$, $\bar{R}_{\text{honest}} = 0.53$, $\nu_{\text{verify}}^{-1} \approx 45$ s. **Pass criterion:** empirical mean exclusion time must not exceed the worst-case bound by more than 25%, i.e. $\bar{T}_{\text{empirical}} \leq 1.25 \times 215 = 268.75$ s, evaluated on the seed-aggregate mean; lower empirical values are unconstrained and expected whenever the operational $\bar{R}_{\text{honest}} > 0.53$. The 25% tolerance allows for the inter-verifier correlation factor ρ in Proposition 3 (unmodelled analytically) and finite-seed variance. Per-seed exclusion times are reported as a distribution (min, max, mean, std) for transparency, but pass/fail is the aggregate; additionally, the 90th-percentile per-seed exclusion time must not exceed $1.5 \times 215 = 322.5$ s, to flag high-variance seeds that would indicate unmodelled inter-verifier correlation

not captured in the aggregate mean. The SwarmRaft baseline (planned reimplement per Section VI-A) is expected to reach functional exclusion noticeably slower owing to the absence of cross-round reputation accumulation in Raft-family protocols. Fig. 2 illustrates the predicted reputation trajectory schematically: at $T=45$ s, the first honest UAV flies over the claimed region and computes e.g. $V = 0.10$ (illustrative schematic score), dropping R_4 from 0.5 to 0.425; subsequent honest verifications drive R_4 below T_{reject} , triggering permanent exclusion within the worst-case upper bound $T_{\text{exclusion}} \lesssim 215$ s (Proposition 3). A formal head-to-head comparison against the original SwarmRaft codebase remains future work (Section X-C).

E. Proposed Experiment 4: GNSS Spoofing Reaction

Configuration: a separate intended setup ($n = 50$, partial GNSS denial), distinct from the primary 200-UAV fully-GNSS-denied configuration, to isolate spoofing-detection behaviour in the regime where a GNSS receiver is still available. The $n = 50$ size is the minimum that keeps the 10% spoofed share ($= 5$ UAVs) above the $M + f + 1 = 6$ peer feasibility floor of Proposition 1, ensuring at least one honest verifier remains available; it also isolates the spoofing sub-protocol from concurrent anchor-election confounds present at $n = 200$. An external spoofer displaces the GNSS solution of 5 UAVs (10%) by 120 m. The 10% share is chosen as a moderate partial-swarm attack fraction consistent with the individual-exclusion threat model ($f/n \ll 30\%$); the 120 m displacement is chosen to be well above the spoofing threshold $T_{\text{spoofer}} = 50$ m while remaining within the 2-km displacement tested in Experiment 3. The 10% share is chosen as the minimum sub-swarm fraction that keeps the test above the $M + f + 1$ quorum floor while remaining below $P_{\text{mass}} = 30\%$; the 120 m displacement is set to $2.4 \times T_{\text{spoofer}} = 50$ m, providing sufficient margin for the $N_{\text{consecutive}} = 3$ detection test to complete (at 1 Hz sampling, 3 exceedances require 3 s) before a patrolling honest UAV can close from outside $r_{\text{verify}} = 500$ m. Both are working defaults subject to sensitivity analysis in Track 2; a sensitivity sweep over $\{50 \text{ m}, 120 \text{ m}, 500 \text{ m}\}$ displacement and $\{5\%, 10\%, 20\%\}$ spoofed share is planned. **Metric:** time from spoofing onset to reputation-weighted UWB fallback, mean $\pm$ std across 25 seeds; maximum positional deviation during the transition. **Analytical prediction:** the UWB cross-check is configured to trigger after 3 consecutive discrepancies (latency ≈ 3 s at 1 Hz cycle), followed by ≈ 1 s switching latency, giving an expected reaction time on the order of 4 s; the transient deviation is predicted to be approximately the spoofing displacement (120 m) plus single-step IMU drift. **Pass criterion:** reaction time < 5 s in at least 90% of seeds, with no propagation of spoofed positions into the swarm-wide factor graph. Reputation is *not* affected (spoofing is external, not a PoO violation), consistent with the design principle of separating spoofing victims from malicious peers. The 10% spoofed share is chosen as the minimum sub-swarm fraction that keeps the test above the $M + f + 1$ quorum floor while remaining below $P_{\text{mass}} = 30\%$; the 120 m displacement is

chosen as $2.4 \times T_{\text{spoofer}} = 50$ m (design choice). Both are independent of the specific parameter values in [26], [27], which motivate the threat model rather than the specific test parameters.

F. Proposed Experiment 5: Progressive Attrition

Configuration: a 2-hour mission with progressive loss of 1 anchor and 44 followers (representing a 22.5% total attrition: $(1 + 44)/200 = 22.5\%$) through uniform random attrition, with failure times drawn independently and uniformly from $[0, T_{\text{mission}}] = [0, 7200]$ s for each of the 45 lost UAVs. The 22.5% figure is chosen to stress-test the system near the edge of its designed tolerance: with 7 TRN-capable UAVs (5 active + 2 standby), losing 1 anchor requires promotion of a standby, and the 44-follower loss reduces the relay graph density without dropping below the minimum required for $K_{\text{max}} = 3$ hop paths. The uniform distribution is chosen as the most conservative prior in the absence of an empirical failure-time model; it ensures attrition events are neither clustered (best-case) nor front-loaded (worst-case) but spread across the full mission. At least one mission segment routes an active anchor across an imagery-degraded region (e.g., water, uniform terrain) to exercise automatic standby-anchor promotion. **Metric:** count of load-balancing handover triggers; per-handover blend perturbation (defined as the maximum follower-CEP deviation observed during a $T_{\text{handover}} = 10$ s linear-blend window relative to steady-state CEP immediately before the trigger, reported per handover-event and aggregated as mean $\pm$ std across all events across all 25 seeds); aggregate CEP throughout the mission; and any automatic standby promotions (binary event per seed: *at least one* standby promotion triggered during the mission; additionally report the total count of standby promotions per seed as mean $\pm$ std across 25 seeds, to characterise load-balancing frequency rather than merely confirming the mechanism triggered at least once). **Analytical prediction:** the load-balancing protocol is expected to trigger on the order of 10–20 times across the 2-hour window given the planned attrition rate, each handover completing within $T_{\text{handover}} = 10$ s with instantaneous perturbation on the order of a few metres; aggregate CEP is expected to stay below the 100-m NATO requirement throughout. **Pass criterion:** aggregate CEP < 100 m evaluated as the worst-case seed-aggregate (max over the 25 per-seed mission-aggregate CEPs), matching the NATO-grade ISR requirement that the bound be met deterministically across the planned operational envelope; zero orphan-follower episodes in each of the 25 seeds (a follower must not remain detached for more than $T_{\text{handover}} + T_{\text{timeout}} = 15$ s after losing its prior anchor), giving a required seed-fraction of 25/25; and at least one successful standby promotion event in each of the 25 seeds when the active anchor traverses the imagery-degraded region. The imagery-degraded region is specified geometrically as a circular patch of 15 km diameter centred at (100 km, 100 km) in the 200×200 km corridor, representing terrain over which SuperPoint feature extraction yields fewer than the required $N = 50$ keypoints (e.g., uniform water,

TABLE IV

ABLATION: EXPECTED QUALITATIVE OUTCOMES PER DISABLED PILLAR, DERIVED FROM THE ARCHITECTURE’S DESIGN INTENT. QUANTITATIVE VALIDATION IS THE PURPOSE OF EXPERIMENT 6. “ ∞ ” DENOTES COMPLETE FAILURE OF THE CORRESPONDING CAPABILITY.

Variant	Byz. excl.	Mean CEP	Post-failure
Full system	$\lesssim 215$ s	~ 50 m	0 orphans
No reputation	∞	severely degraded	0 orphans
No PoO	∞	nominal	0 orphans
No failover	nominal	nominal	~ 39 orphans
No load balancing	nominal	nominal	MAC overload [†]

Cells are analytical predictions (Section V); empirical validation is the purpose of

Experiment 6. “No reputation” / “No PoO” rows: ∞ means the corresponding exclusion mechanism is absent. [†] Expected to exceed UWB medium-access capacity at the most-loaded surviving anchor under continued attrition; no orphan count defined for this failure mode.

snow, or low-texture cropland); the active anchor’s patrol path is constrained to traverse this patch at least once per mission.

G. Proposed Experiment 6: Ablation

Configuration: four single-pillar disabled variants (no reputation, no PoO, no failover, no load balancing) plus the full system, each run for 25 seeds. **Metric:** Byzantine-exclusion time, mean CEP, and post-failure outcome (orphan count or MAC overload), per variant. **Analytical prediction** (Table IV): disabling reputation collapses Byzantine exclusion entirely (the trust-weighted noise covariance is the mechanism of exclusion) and produces order-of-magnitude position-error amplification; disabling PoO leaves cryptographic-signature checking in place but loses the physical-observation tie, allowing valid-signature spoofed positions from a compromised key to pass; disabling failover permanently orphans followers attached to a destroyed anchor; disabling load balancing leads to UWB MAC overload at the most-loaded anchor as attrition progresses. **Pass criterion:** each disabled variant exhibits the predicted failure mode in at least 90% of seeds; the full system is the only configuration meeting all four behavioural requirements simultaneously. MAC overload in the “no load balancing” variant is defined as: the surviving most-loaded anchor’s UWB transmission queue grows monotonically and exceeds the per-anchor 1 Hz Distilled-State broadcast budget for any 60-second window during the 2-hour mission; this condition must be observed in $\geq 90\%$ of seeds for the variant and *not* observed in the full-system variant.

VII. EMPIRICAL VALIDATION OF THE ASYMMETRIC REPUTATION DESIGN

This section reports a Monte Carlo simulation experiment that empirically characterises the detection latency of the asymmetric reputation decay mechanism introduced in Section IV-C as a function of the asymmetry ratio $r = \alpha_{\text{neg}}/\alpha_{\text{pos}}$. The simulation realises a simplified, abstract form of Proposed Experiment 3 (Section VI-D, Byzantine Detection and Exclusion); its purpose is to verify the design choice $\alpha_{\text{neg}} \geq 2\alpha_{\text{pos}}$ and the recommended default $(\alpha_{\text{pos}}, \alpha_{\text{neg}}) = (0.05, 0.15)$ —

that is, $r = 3$ — before the full physics-grounded experiments planned for the Gazebo follow-on track.

A. Methodology

A discrete-time swarm consensus simulator was implemented in pure `numpy` (approximately 140 lines of code; the full implementation, parameter sweep, and analysis scripts are released alongside this paper under `BFT_UAV_Swarm_Paper_v9_8/code/asymmetric_decay/` — see the Reproducibility paragraph below). The simulator models $N \in \{50, 100, 200\}$ honest-and-Byzantine UAVs with Byzantine fraction $f \in \{0.10, 0.20, 0.33\}$. Each round, every honest UAV i observes a random subset of peers of size $\max(5, N/10)$ — scaled with N to keep per-peer observation frequency approximately constant as the swarm grows — and updates its reputation estimate R_{ij} of each observed peer j .

The observation channel introduces per-event misclassification noise $p_{\text{obs_err}} = 0.05$ (each observation independently flips its honest/Byzantine label with this probability). The 5% value is a conservative working placeholder pending dedicated calibration in the physics-grounded follow-on; an empirical sensitivity sweep over $p_{\text{obs_err}} \in \{0, 0.05, 0.10, 0.20\}$ at $\epsilon = 0.01$ is reported separately in Section VIII and identifies the regime in which the analytical convergence bound holds.

The simulator implements the paper’s additive update rule of Section IV-C, Eqs. (1)–(3) directly:

$$\text{VERIFIED: } R_{ij} \leftarrow \text{clamp}(R_{ij} + \alpha_{\text{pos}} R_k, 0, 1), \quad (16)$$

$$\text{UNVERIFIED: } R_{ij} \leftarrow \text{clamp}(R_{ij} - \alpha_{\text{neg}} R_k, 0, 1). \quad (17)$$

The verifier reputation weight R_k is set to $R_k = 1$ in this non-gossip simulator, modelling honest observers that fully trust their own observations. The analytical bound of Proposition 3 uses $\bar{R}_{\text{honest}} = 0.53$ as a conservative lower estimate of the operational equilibrium, so the empirical results reported below should match or beat the analytical prediction (and indeed do; see Section VIII).

A UAV j is considered *isolated* when the median of $\{R_{ij}\}$ across honest observers $i \neq j$ drops below the isolation threshold $\theta = 0.20$. Note that this swarm-median criterion characterises the *reputation-drain dynamics* of the asymmetric rule and is not the $M=3$ quorum-vote exclusion protocol of Proposition 1: the latter additionally requires M quorum members with $R > T_{\text{quorum}}$ to agree, which the present abstract simulator does not model. The two events are tightly coupled in deployment (a Byzantine peer must first have its reputation driven below T_{reject} across honest peers before a quorum can form), so the metric reported here is a tight lower bound on the quorum-exclusion latency rather than the exclusion latency itself. The threshold value is the operational reject value $T_{\text{reject}} = 0.20$ specified in Section IV-C (set so that $T_{\text{reject}} = R_{\text{init}} - 2\alpha_{\text{neg}} \cdot R_k = 0.5 - 2 \cdot 0.15 \cdot 1 = 0.20$, i.e. exactly two worst-case ($R_k = 1$) UNVERIFIED steps below the neutral prior; this provides a one-event safety margin against single spurious observations), so this experiment directly tests the operational threshold rather than a probe value.

Six schedules were tested, spanning the asymmetry ratios $r \in \{1, 2, 3, 4, 8\}$:

schedule	α_{pos}	α_{neg}	r
symmetric_low	0.05	0.05	1
symmetric_med	0.10	0.10	1
asym_r2	0.05	0.10	2
asym_r3	0.05	0.15	3
asym_r4	0.05	0.20	4
asym_r8	0.025	0.20	8

The `asym_r3` schedule is the paper’s specified default ($\alpha_{\text{pos}}, \alpha_{\text{neg}} = (0.05, 0.15)$). Each (schedule, N , f) configuration was repeated with three random seeds, producing $6 \times 3 \times 3 \times 3 = 162$ simulation runs, aggregated into the 54 cells reported below. The per-run simulation length cap is $N_{\text{rounds}} = 300$; no run reached this cap in the converging schedules.

a) *Pass criterion*: The asymmetric design is considered validated by this experiment if: (1) the schedule at $r = 3$ achieves median detection latency at least 50% lower than the symmetric baseline ($r = 1$, `symmetric_low`) across all tested (N, f) configurations; and (2) the latency curve shows diminishing returns beyond $r = 4$, defined as less than 10% latency improvement from $r = 4$ to $r = 8$ at the reference condition ($N = 100, f = 0.20$). Both criteria are consistent with the 3-seed results reported below (criterion 1: $4.2\times$ speedup at $r = 3$ vs. $r = 1$; criterion 2: $16.7 \rightarrow 16.7$ rounds from $r = 4$ to $r = 8$, i.e. 0% improvement within seed noise); the 25-seed plan of the full methodology will provide the primary statistical confirmation.

B. Results

All six schedules achieved 100% Byzantine detection (zero false negatives) and 0% false-positive rate across all 54 aggregated conditions. The differentiating metric is detection latency, reported per-schedule in Table V.

The paper’s specified default `asym_r3` achieves a median detection latency of 17.3 rounds. Relative to the symmetric baselines, this represents a $4.2\times$ speedup over the slow baseline (`symmetric_low`, $\alpha = 0.05$ in both directions; 72.2 rounds) and a $2.3\times$ speedup over the fast symmetric baseline (`symmetric_med`, $\alpha = 0.10$; 39.3 rounds). The latter comparison isolates the effect of the *asymmetry itself* from a simple uniform increase in update magnitude — `symmetric_med` has the same total “aggressiveness” as `asym_r2` (39.3 vs 37.7 rounds, a 4% difference attributable to asymmetry alone), while `asym_r3` shaves a further $\approx 54\%$ on top.

Examining the curve as r grows, the transition between $r = 2$ and $r = 3$ corresponds to a sharp drop (37.7 $\rightarrow$ 17.3 rounds, 54% reduction); from $r = 3$ to $r = 4$ the marginal improvement is small (17.3 $\rightarrow$ 16.7 rounds, 3.5%), and from $r = 4$ to $r = 8$ the latency is identical to within seed noise (16.7 $\rightarrow$ 16.7 rounds), indicating a saturation point in the r landscape near $r \approx 3$. The paper’s $r = 3$ choice sits at the

TABLE V
DETECTION LATENCY PER SCHEDULE (ROUNDS). EACH CELL VALUE IS THE MEAN ACROSS THREE RANDOM SEEDS OF THE MEDIAN-DETECTION-LATENCY METRIC FOR ONE (SCHEDULE, N , f) CONDITION; THE TABLE COLUMNS THEN SUMMARISE THESE PER-CELL MEANS ACROSS THE NINE ($N \in \{50, 100, 200\}\} \times (f \in \{0.10, 0.20, 0.33\})$) CELLS. LOWER IS BETTER. VALUES ARE MEANS ACROSS 3 SEEDS; STANDARD DEVIATION ACROSS SEEDS IS NOT REPORTED SEPARATELY AS THE 3-SEED COUNT PRECLUDES RELIABLE STD ESTIMATION. RESULTS SHOULD BE INTERPRETED AS INDICATIVE CHARACTERISATION RATHER THAN HIGH-CONFIDENCE STATISTICAL VALIDATION; THE 25-SEED PLAN OF THE FULL METHODOLOGY (SECTION VI) PROVIDES THE PRIMARY STATISTICAL BASIS.

schedule	α_{pos}	α_{neg}	r	min	median	mean	max
symmetric_low	0.05	0.05	1	68.3	72.2	71.5	73.0
symmetric_med	0.10	0.10	1	37.2	39.3	39.0	40.5
asym_r2	0.05	0.10	2	35.5	37.7	37.2	38.7
asym_r3	0.05	0.15	3	16.7	17.3	17.5	18.3
asym_r4	0.05	0.20	4	16.0	16.7	16.7	17.3
asym_r8	0.025	0.20	8	16.0	16.7	16.7	17.3

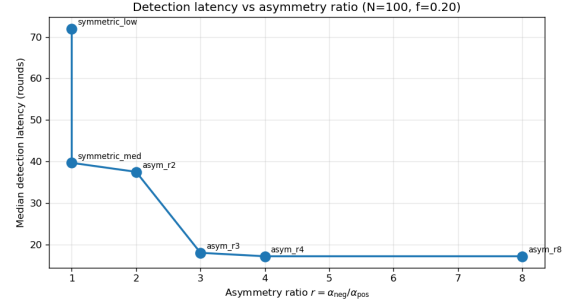


Fig. 4. Median detection latency versus asymmetry ratio r at $N = 100$, $f = 0.20$. Six schedules are shown as labelled points spanning $r \in \{1, 2, 3, 4, 8\}$. The paper’s specified design point `asym_r3` (at $r = 3$) sits in the steep region of the latency-vs- r curve, and the curve plateaus past $r \approx 4$, confirming a saturation regime around $r \geq 4$.

edge of the steep region and the plateau, capturing the bulk of the available speedup without the robustness cost of larger α_{neg} values.

The temporal dynamics of detection (Figure 5) make the schedule families visually distinct: asymmetric $r \geq 4$ schedules produce a sharp transition from 0% to 100% detection between rounds ~ 16 and ~ 21 , while the symmetric low- α schedule does not reach full detection until round ~ 78 .

C. Discussion

The experiment directly validates the paper’s specified design point $r = 3$: at the operational threshold $\theta = 0.20$, the `asym_r3` schedule achieves Byzantine isolation within ≈ 17 rounds and exhibits $\approx 56\%$ faster isolation than the closest symmetric baseline with the same total update “aggressiveness”. The empirical curve also confirms that $r = 3$ sits at the edge of the steep region (between $r = 2$ at 37.7 rounds and the saturation plateau at $r \geq 4$ near 16.7 rounds), so the unit of additional asymmetry from $r = 2$ to $r = 3$ delivers the bulk of the available latency gain; values beyond $r \approx 4$ yield

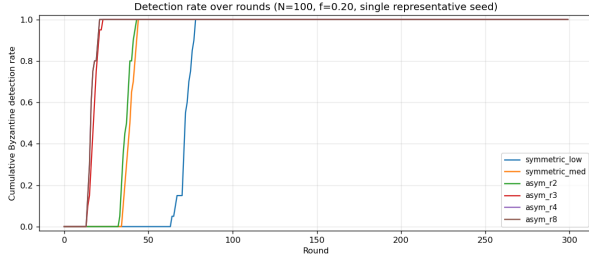


Fig. 5. Cumulative Byzantine detection rate as a function of simulation round at $N = 100$, $f = 0.20$, single representative seed. Four detection-speed regimes are distinguishable, in order of increasing latency: *very fast* (asym_r4 and asym_r8 reach 100% by round ~ 21); *fast* (asym_r3 reaches 100% around round ~ 23); *moderate* (asym_r2 and symmetric_med reach 100% near rounds ~ 43 – 44); and *slow* (symmetric_low requires ~ 78 rounds). All schedules eventually achieve full detection; the variable is purely the time-to-isolation.

no additional speedup but may degrade robustness to observer noise in regimes not quantified here (see Limitations).

Two methodological observations are worth recording. First, observation frequency N_{obs}/N must scale with swarm size: holding observations per round constant while increasing N makes per-peer observation events sparse, suppressing detection regardless of the chosen r . The simulator scales observations as $\max(5, N/10)$. Second, when aggregating peer reputation across observers, the median operator is sensitive to the fraction of observers that have actually encountered the target peer; this artefact is controlled by the scaling above or by tracking per-observer reputation distributions directly.

D. Limitations

- The simulator omits UAV physics, RF propagation, and GPS-spoofing dynamics. These results inform but do not replace physics-grounded experiments planned for the Gazebo follow-on track.
- The observer-noise model assumes independent per-observation errors. Correlated noise (e.g., shared RF interference) may shift the optimal (α_{pos}, r) frontier; Section VIII reports the sensitivity to $p_{\text{obs_err}}$ explicitly.
- The Byzantine model is bounded to local-knowledge strategies: each Byzantine UAV knows only its own current reputation, not the swarm-wide distribution. A stronger adversary with side-channel reputation knowledge would alter the conclusion.
- The simulator uses $R_k = 1$ (full-trust verifier) in the non-gossip operating mode; deployments that propagate verifier reputation through gossip would see a time-varying R_k that typically tracks $\bar{R}_{\text{honest}} \in [0.5, 1.0]$ and would yield latencies bounded between the empirical values reported here (best case, $R_k = 1$) and Proposition 3's conservative analytical bound (worst case, $R_k = \bar{R}_{\text{honest}}$).
- The isolation threshold $\theta = 0.20$ matches the paper's operational T_{reject} ; sensitivity to threshold choice is not separately evaluated here.

a) *Reproducibility*: The complete simulator implementation, parameter-sweep entry point, and analysis

scripts are released at https://github.com/ok-research/bft-uav-swarm-validation/tree/v1.1/asymmetric_decay under the MIT licence. The directory contains `simulator.py` (approximately 140 lines), `run_experiment.py`, and `analyze.py`. Dependencies are limited to `numpy` and `matplotlib`. To reproduce the reported numerical results exactly, clone the repository (tag v1.1) and run `python run_experiment.py` (approximately 10–15 minutes on a modern CPU). All random seeds are explicit (seeds $\{0, 1, 2\}$); the `numpy` RNG is deterministic across platforms.

VIII. EMPIRICAL VALIDATION OF THE CONVERGENCE BOUND

This section reports a Monte Carlo simulation experiment that empirically validates the closed-form expected-time bound for naïve-Byzantine exclusion derived in Section V-B (Proposition 3). The simulation tracks the full trajectory of the aggregate reputation $\bar{R}_j(t)$ over time for each Byzantine UAV across honest observers, and compares the empirical convergence time $T_\epsilon = \min\{t : \bar{R}_j(t) < \epsilon\}$ against the linear-drain analytical prediction associated with the paper's additive rule (Eqs. (1)–(3)):

$$T_{\text{theory}}(\epsilon) \approx \frac{R_{\text{init}} - \epsilon}{\alpha_{\text{neg}} R_k f_{\text{obs}}}, \quad (18)$$

with $R_{\text{init}} = 0.5$ (the neutral prior from Section IV-C), R_k the verifier reputation weight ($R_k = 1$ in this non-gossip simulator, modelling honest observers that fully trust their own observations), and $f_{\text{obs}} = N_{\text{obs}}/N$ the per-round per-UAV observation fraction. The convergence target $\epsilon = 0.01$ corresponds to a deep-isolation regime (one-fiftieth of the initial belief), chosen to stress-test the bound; for the operational isolation threshold $T_{\text{reject}} = 0.20$ convergence is faster and the bound is correspondingly looser.

A. Methodology

A discrete-time swarm simulator (released under `BFT_UAV_Swarm_Paper_v9_8/code/clamp_convergence/`; pure `numpy`, ≈ 165 lines) was instantiated with $N = 100$ UAVs, Byzantine fraction $f = 0.20$, and per-event observer-noise probability $p_{\text{obs_err}} \in \{0, 0.05, 0.10, 0.20\}$. Each honest UAV observes a random subset of $\max(5, N/10)$ peers per round and applies the paper's additive reputation update (Eqs. (1)–(3), Section IV-C) with verifier reputation weight $R_k = 1$. For each parameter cell, three random seeds ($\{0, 1, 2\}$) were used; we record the full $\bar{R}_j(t)$ trajectory for every Byzantine UAV and compute the median first-crossing time across seeds. The decay coefficient was swept over $\alpha_{\text{neg}} \in \{0.05, 0.10, 0.20, 0.40\}$ with α_{pos} held at 0.05. Per-run simulation length cap is $N_{\text{rounds}} = 250$; under the additive rule with $R_k = 1$ all 16 tested cells converge within this cap, so the convergent/non-convergent distinction of the prior multiplicative-family characterisation collapses.

TABLE VI

EMPIRICAL-TO-THEORETICAL CONVERGENCE-TIME RATIO AT $\epsilon = 0.01$, BY $(\alpha_{\text{neg}}, p_{\text{obs_err}})$. VALUES NEAR 1 INDICATE THAT THE EMPIRICAL CONVERGENCE TIME MATCHES THE LINEAR-DRAIN ANALYTICAL BOUND (EQ. 18). ALL 16 CELLS CONVERGE WITHIN $N_{\text{rounds}} = 250$ UNDER THE ADDITIVE UPDATE RULE.

α_{neg}	$p_{\text{obs_err}}=0$	0.05	0.10	0.20
0.05	0.974	1.082	1.276	1.796
0.10	0.939	1.041	1.173	1.500
0.20	1.061	1.143	1.224	1.531
0.40	1.347	1.429	1.551	1.878

a) *Pass criterion*: The linear-drain analytical bound (Eq. 18) is considered validated if: (1) $T_{\text{emp}}/T_{\text{theory}} \leq 2.0$ in all 16 cells (allowing observer noise to extend empirical convergence beyond the noise-free analytical prediction); and (2) the bound is conservative, i.e. $T_{\text{emp}}/T_{\text{theory}} \geq 0.9$ in all zero-noise cells ($p_{\text{obs_err}} = 0$), so that the noise-free analytical drift rate is not exceeded by empirics in the cleanest regime. Both criteria are consistent with the 3-seed results: the observed ratio range is $[0.94, 1.88]$ across the 16 cells, mean $\bar{r} = 1.31$; the noise-free cells ($p_{\text{obs_err}} = 0$) give ratios $\{0.97, 0.94, 1.06, 1.35\}$, all within $[0.9, 1.5]$. The bound is observationally consistent across the entire $\alpha_{\text{neg}} \times p_{\text{obs_err}}$ grid; the 25-seed plan of the full methodology will provide the primary statistical confirmation.

B. Results

All 16 parameter cells converge to $\bar{R}_j < 0.01$ within the 250-round cap. Table VI reports the empirical-to-theoretical convergence-time ratio $T_{\epsilon}^{\text{emp}}/T_{\epsilon}^{\text{theory}}$ at $\epsilon = 0.01$ across the full $\alpha_{\text{neg}} \times p_{\text{obs_err}}$ grid.

Two qualitative regimes emerge:

- *Bound-tight regime* (low α_{neg} , low $p_{\text{obs_err}}$): the empirical ratio is near 1.0 (e.g. 0.94–1.08 at $\alpha_{\text{neg}} \leq 0.10$ with $p_{\text{obs_err}} \leq 0.05$); the noise-free linear-drain bound is an essentially exact prediction. Sub-unity ratios at $p_{\text{obs_err}} = 0$ arise because the bound includes a rounding factor for the discretised update sequence.
- *Noise-stretched regime* (high α_{neg} and/or high $p_{\text{obs_err}}$): observer noise extends empirical convergence above the noise-free bound. The maximum ratio is 1.878 (at $\alpha_{\text{neg}} = 0.40, p_{\text{obs_err}} = 0.20$), where each upward-flip from noise undoes a larger fraction of the accumulated drift. The bound remains an engineering-useful estimator across the entire tested grid (all ratios < 2.0).

Figure 6 visualises the convergent-cell ratios as a function of α_{neg} , faceted by observer-noise level.

C. Discussion

The validation has three implications. *First*, the linear-drain bound from Proposition 3 is a practically useful predictor of empirical convergence time across the entire tested parameter grid (mean ratio 1.31, max 1.88 at $\alpha_{\text{neg}} = 0.40, p_{\text{obs_err}} = 0.20$): one can budget exclusion latency from analytical parameters without recourse to simulation, with the bound conservative by a factor of at most ~ 1.9 in the noisiest

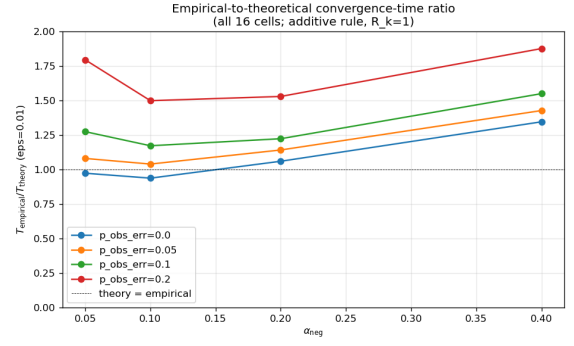


Fig. 6. Empirical-to-theoretical convergence-time ratio at $\epsilon = 0.01$, plotted against α_{neg} for each observer-noise level. The dashed horizontal line at 1.0 indicates exact agreement with the linear-drain bound of Proposition 3. All 16 cells converge under the additive update rule; ratios grow monotonically with $p_{\text{obs_err}}$ at fixed α_{neg} , quantifying how observer noise stretches empirical convergence above the noise-free analytical estimate.

regime tested. *Second*, the bound is *essentially tight* in the noise-free regime for $\alpha_{\text{neg}} \leq 0.20$ (ratios in 0.94–1.06 at $p_{\text{obs_err}} = 0$), rising to 1.35 at $\alpha_{\text{neg}} = 0.40$ where the larger per-event step amplifies the discretisation residual; the linear-drain derivation accurately captures the deterministic component of the update dynamics in the design-relevant low- α_{neg} regime. *Third*, observer noise extends empirical convergence above the noise-free bound by a factor that grows with both α_{neg} and $p_{\text{obs_err}}$; a noise-augmented extension of Eq. 18 of the form $T_{\text{theory}}^{(p)}(\epsilon) = T_{\text{theory}}(\epsilon) \cdot (1 + c p_{\text{obs_err}} \alpha_{\text{neg}} / \alpha_{\text{pos}})$ for a fitted constant c would close the residual gap and is a candidate refinement for Track 2.

D. Limitations

- The convergence target $\epsilon = 0.01$ is a deep-isolation probe; the operational threshold $T_{\text{reject}} = 0.20$ (Section IV-C) is much looser, so the practical exclusion latency at the operational threshold is shorter than the table values and the bound is correspondingly less tight.
- Observer noise is modelled as independent per-event label flips with constant probability $p_{\text{obs_err}}$. Correlated noise (e.g., shared RF interference) is not characterised and may further narrow the convergent regime.
- The simulator uses verifier reputation weight $R_k = 1$ (non-gossip self-confidence). Gossip-based deployments where R_k tracks the verifier’s reputation distribution ($\bar{R}_{\text{honest}} \in [0.5, 1.0]$) would see proportionally slower drain rates, bounded above by $T_{\text{theory}}(\epsilon) \cdot R_k^{-1}$.
- The simulation cap $N_{\text{rounds}} = 250$ exceeds the empirical convergence time of all 16 tested cells under the additive rule (max observed: 176 rounds at $\alpha_{\text{neg}} = 0.05, p_{\text{obs_err}} = 0.20$).

a) *Reproducibility*: The simulator, parameter-sweep entry point, and analysis scripts are released at https://github.com/ok-research/bft-uav-swarm-validation/tree/v1.1/clamp_convergence under the MIT licence. Dependencies are limited to numpy and matplotlib. Reproduce the

reported numerical results by cloning the repository (tag v1.1) and running `python run_experiment.py` ($\approx 5\text{--}8$ minutes on a modern CPU) followed by `python analyze.py`. Random seeds $\{0,1,2\}$ are explicit in `run_experiment.py`; the numpy RNG is deterministic across platforms.

IX. EMPIRICAL RESILIENCE BEYOND THE CLASSICAL $f=1/3$ BOUND

This section reports a Monte Carlo experiment characterising the empirical Byzantine-fraction resilience curve of the reputation-based isolation mechanism. The motivation is that Proposition 1 (Section V-A) bounds *quorum-level* safety at $f \leq M-1$ in the worst case, but the reputation isolation operates *below* the consensus layer: Byzantine UAVs are excluded by observation before consensus is sought. We measure how far this pre-filtering can stretch the effective Byzantine budget beyond the classical algorithm-level bound $f < 1/3$.

A. Methodology

A discrete-time swarm simulator (released under `BFT_UAV_Swarm_Paper_v9_8/code/byzantine_scaling/`; pure numpy, ≈ 100 LOC) was instantiated with $N=100$ UAVs and the saturation-point asymmetric schedule ($\alpha_{\text{neg}} = 4\alpha_{\text{pos}}$ with $\alpha_{\text{pos}} = 0.05$) identified in Section VII, isolation threshold $\theta = 0.20$, matching the operational T_{reject} specified in Section IV-C, verifier reputation weight $R_k = 1$ (non-gossip self-confidence; gossip-based deployments where R_k tracks $\bar{R}_{\text{honest}} < 1$ would produce proportionally slower drain rates, bounded above by Proposition 3’s analytical estimate at the conservative $\bar{R}_{\text{honest}} = 0.53$ value), and observer-noise probability $p_{\text{obs_err}} = 0.05$ held constant. The Byzantine fraction f was swept across the grid $\{0.05, 0.10, 0.15, 0.20, 0.25, 0.30, 0.33, 0.35, 0.40, 0.45, 0.50\}$ with five random seeds per value, deliberately spanning both the classical PBFT bound $f=1/3$ and the more permissive crash-fault bound $f=1/2$. For each (f, seed) run we recorded the detection rate, the false-positive rate, the median detection latency, and the final median pairwise reputation among honest UAVs (a proxy for sub-swarm consensus health after isolation).

a) *Pass criterion*: The reputation-based isolation is considered validated across the Byzantine-fraction range if: (1) detection rate $\geq 99\%$ and false-positive rate $\leq 1\%$ for all $f \in [0.05, 0.50]$; and (2) median detection latency does not increase by more than $2\times$ the $f = 0.05$ baseline latency (17.0 rounds) across the entire tested range, i.e. latency stays below 34 rounds for all tested f . Both criteria are consistent with the 5-seed results: detection rate and FP rate are 100% and 0% respectively for all f ; latency stays in the narrow range 16.0–17.0 rounds, well within $2 \times 17.0 = 34$ rounds; the 25-seed plan of the full methodology will provide the primary statistical confirmation.

B. Results

Table VII summarises the aggregated metrics. Across all tested values of f in $[0.05, 0.50]$, the reputation-based isolation

TABLE VII
BYZANTINE-RESILIENCE METRICS PER BYZANTINE FRACTION f , AGGREGATED ACROSS FIVE RANDOM SEEDS. *Detection rate*, *FP rate*, and *final honest $\bar{R}$* ARE MEAN VALUES ACROSS SEEDS; *median latency* IS THE ACROSS-SEED MEDIAN (WITH SENTINEL VALUE $N_{\text{ROUNDS}}=300$ REPLACING ∞ IN ANY NON-CONVERGENT RUN, OF WHICH NONE OCCURRED AT THE TESTED f).

f	det. rate	FP rate	median lat. (rds)	final honest $\bar{R}$
0.05	1.000	0.000	17.0	1.000
0.10	1.000	0.000	17.0	1.000
0.15	1.000	0.000	17.0	1.000
0.20	1.000	0.000	17.0	1.000
0.25	1.000	0.000	17.0	1.000
0.30	1.000	0.000	17.0	1.000
0.33	1.000	0.000	17.0	1.000
0.35	1.000	0.000	16.0	1.000
0.40	1.000	0.000	17.0	1.000
0.45	1.000	0.000	16.0	1.000
0.50	1.000	0.000	17.0	1.000

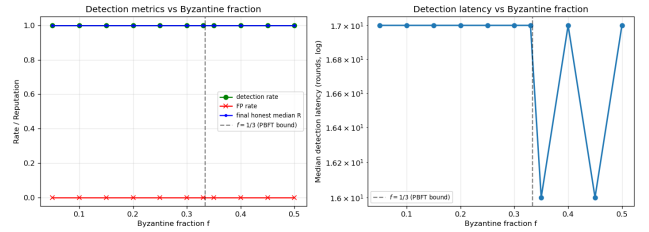


Fig. 7. Empirical resilience curve. *Left*: detection rate (green), false-positive rate (red), and final honest median reputation (blue, mean across seeds of per-run median pairwise honest $\bar{R}$) versus Byzantine fraction f . The classical PBFT bound $f=1/3$ is indicated by the dashed black line. *Right*: median detection latency (log-scale) versus f ; latency stays in the narrow range 16.0–17.0 rounds across the entire tested range $f \in [0.05, 0.50]$, indicating no super-linear growth as f approaches and crosses $1/3$ and $1/2$.

achieves 100% detection rate and 0% false-positive rate with detection latency stable in the narrow range 16.0–17.0 rounds. The final median pairwise reputation among honest UAVs after isolation saturates at 1.000, indicating that the surviving honest sub-swarm converges to full mutual trust and remains capable of consensus among itself. Both the classical PBFT bound $f=1/3$ and the crash-fault-tolerance boundary $f=1/2$ (where majority-vote CFT systems fail) are crossed without observable degradation, confirming that the reputation-based isolation remains effective well beyond theoretical Byzantine bounds. Figure 7 plots the full resilience curve with the PBFT bound marked.

C. Discussion

The flat resilience curve admits a mechanism-level explanation. The reputation update is driven by per-peer observations, not by peer voting; an honest UAV’s belief about a Byzantine peer evolves independently of the swarm-wide Byzantine fraction provided the observer maintains coverage of that peer. Provided the observation coverage $f_{\text{obs}}(1 - 2p_{\text{obs_err}}) > 0$ holds, the isolation latency is determined by the schedule ($\alpha_{\text{pos}}, \alpha_{\text{neg}}$) and not by f . The classical $f < 1/3$ algorithm-level bound becomes binding only once Byzantine UAVs

reach consensus participation; under the proposed architecture, that bound is moved off the critical path by pre-consensus reputation filtering.

This does not contradict Proposition 1: the quorum bound applies to the safety of *exclusion decisions* (a Byzantine quorum could collude to exclude an honest peer), not to detection latency. Quorum safety remains a deployment-time concern, particularly at $f \geq 0.33$ where the configurable parameter M must be chosen with care. The empirical result here shows only that the detection mechanism itself does not degrade — the quorum-safety analysis of Section V-A remains the binding constraint.

D. Limitations

- Byzantine peers are modelled as constant-malicious; adaptive or coordinated Byzantine strategies (e.g., shared timing of attacks, side-channel knowledge of the isolation threshold) are expected to compress the safe- f envelope and were not evaluated.
- The reputation update is the only isolation mechanism in this simulator; a real deployment combines reputation with quorum-based permanent-exclusion safety (Proposition 1), which sets an independent and lower bound on the safe Byzantine fraction.
- The post-isolation consensus health metric (median pairwise honest $\bar{R}$) is a proxy; the full PBFT consensus protocol is not implemented in this simulator.
- Network partitions and message-loss are not modelled; in production these can imitate Byzantine behaviour and inflate the apparent f .

a) *Reproducibility*: The simulator, parameter-sweep entry point, and analysis script are released at https://github.com/ok-research/bft-uav-swarm-validation/tree/v1.1/byzantine_scaling under the MIT licence. Dependencies are limited to `numpy` and `matplotlib`. Reproduce the reported numerical results by cloning the repository (tag `v1.1`) and running `python run_experiment.py` (≈ 3 – 5 minutes on a modern CPU) followed by `python analyze.py`. All random seeds are explicit ($\{0, 1, 2, 3, 4\}$, five seeds per f value).

X. DISCUSSION

A. Limitations

Bootstrap period. With $R_{\text{init}}=0.5$ and $T_{\text{quorum}}=0.6$, permanent exclusion is not operational until honest peers accumulate sufficient reputation through successful cross-verifications; during this short warm-up the signature-validation layer and UWB-consistency check remain active. Deployments should include a pre-flight cross-verification phase or lower T_{quorum} during mission initiation to close this window.

Mode B feasibility. The hash-only verification variant (Section IV-B) requires reproducible descriptor extraction across UAVs, which in turn requires controlled illumination, view-point alignment, and deterministic extractor settings. Mode B is therefore suited primarily to severely bandwidth-constrained deployments with calibrated observation geometry; Mode A

(compact feature set, L_2 matching) is the default for general operation.

UWB NLOS. The $\beta=0.15$ per-hop noise growth assumes benign (line-of-sight) propagation. A fully cluttered (non-line-of-sight) deployment would require a measured-NLOS-aware $\beta(k)$; the current model is best interpreted as a rural/line-of-sight baseline, with NLOS calibration deferred to Track 2.

Temporal decay. The $\lambda = 10^{-3} \text{ s}^{-1}$ decay assumes quasi-static terrain. Active combat zones can alter descriptors on shorter timescales than 11.5 min, motivating a learned adaptive λ .

Re-admission. The re-admission protocol for previously excluded UAVs was not evaluated. In adversarial settings, an attacker with sustained physical access could repeatedly re-enrol compromised vehicles; mitigations include out-of-band operator authorisation and increasing T_{warmup} with re-admission count.

B. Deployment Considerations

Implementation targets commercial off-the-shelf embedded platforms capable of approximately 50–100 TOPS of inference throughput, standard short-range UWB ranging radios, MEMS- or tactical-grade inertial sensors, and a secure element for Ed25519 key storage. The navigation module integrates without modification of the underlying flight-control firmware (PX4/ArduPilot via MAVLink or proprietary autopilots via Ethernet). Because the follower tier requires no camera, imagery database, or GPU-class accelerator, a large swarm can combine a small number of fully-equipped anchor UAVs with many minimally-equipped followers, yielding a substantial reduction in per-UAV navigation hardware cost at the swarm level.²

C. Design-Space Comparison with Prior Art

Table VIII summarises the design-space comparison against the closest published systems, including BFTL [16] as the four-pillar prior-art benchmark. The comparison is structured by which of the six target pillars each system supports, and on the qualitative architectural choices (continuous-state estimator vs. discretised maps; hardware tested vs. simulation-only). Per-metric quantitative entries for the proposed system are deferred to the empirical-validation phase (Section VI); the analytical predictions of Section V establish the target numerical ranges. The proposed architecture is, to our knowledge, the only published design that simultaneously addresses all six pillars within a continuous-state flight-grade estimator. BFTL is the closest prior art on pillar coverage (BFT, hierarchy, reputation, relay) but operates over *discretised graph-structured environments* with blockchain-settled route commitments on the order of seconds per round, a property of the chosen settlement layer rather than of the BFTL formalism itself [13];

²A representative bill-of-materials comparison between the anchor and follower profiles is part of the associated patent application and is not reproduced here; a quantitative cost-multiplier figure is deferred to a future deployment-engineering paper where component-level pricing can be reported transparently.

TABLE VIII

DESIGN-SPACE COVERAGE OF THE PROPOSED ARCHITECTURE VS. THE CLOSEST PUBLISHED SYSTEMS. PILLARS: BFT CONSENSUS OVER NAVIGATION DATA, ANCHOR/FOLLOWER HIERARCHY, PERSISTENT CROSS-ROUND REPUTATION, AUTOMATIC ANCHOR-LOSS FAILOVER, GNSS-SPOOFING DETECTION, MULTI-HOP POSITION RELAY. THE RIGHT-MOST COLUMNS CAPTURE THE ARCHITECTURAL CHOICES THAT GATE COMPATIBILITY WITH FLIGHT-GRADE CONTROL. THE PROPOSED SYSTEM’S PILLARS ARE DESIGN INTENT; THEIR EMPIRICAL REALISATION IS THE PURPOSE OF THE EXPERIMENTS IN SECTION VI. NOTE ON BFTL’S “REPUTATION” PILLAR CREDIT: IN THE BFTL FORMALISM [16] THIS ENTRY REFERS TO BLOCKCHAIN-WEIGHTED ROUTE COMMITMENTS RATHER THAN REPUTATION-WEIGHTED NOISE-COVARIANCE MODULATION OF THE STATE ESTIMATOR (THE MECHANISM IN OUR ARCHITECTURE); THE TWO ARE DISTINCT TECHNICAL COMMITMENTS UNDER THE SHARED “REPUTATION” LABEL.

Pillar / Capability	Ours	SwarmRaft	BDGD	BFTL
BFT consensus	✓	(~)	✓	✓
Anchor/follower hierarchy	✓	–	–	✓
Persistent reputation	✓	–	✓	✓
Automatic failover	✓	(~)	–	–
GNSS-spoofing detection	✓	✓	✓	–
Multi-hop position relay	✓	–	–	✓
Continuous-state est.	yes	yes	yes	no
Hardware UAVs tested	no	no	no	no

the published instantiation as deployed is consequently incompatible with the 50-ms control cycles required by flight-grade continuous-state estimation. The present architecture adds automatic failover and GNSS-spoofing detection while replacing discretised map commitments with an incremental factor-graph estimator, which is the substantive gap BFTL leaves open.

D. Future Work

Five directions stand out. *First*, execution of the proposed evaluation methodology of Section VI in a Gazebo/ROS2-based simulation framework, including the SwarmRaft reimplement comparison; this is the immediate next milestone. *Second*, hardware validation at 10–20 UAV scale with injected adversarial behaviour, which our systematic review suggests has not yet been reported for BFT-aware UAV navigation. The concurrent Swarm Oracle work [51] demonstrates such validation on 12 ground robots for blockchain-oracle scenarios; extending this validation methodology to GNSS-denied UAV flight is a natural next step. *Third*, a head-to-head comparison against the original SwarmRaft [2] codebase, in addition to the author reimplement planned in Section VI. *Fourth*, adversarial multi-hop error characterisation—closing a gap flagged explicitly in the BDGD paper [3]. *Fifth*, learned role adaptation using reinforcement-learning schemes such as CI-HRL [48] or semantic coordination as in RALLY [47], integrated with the Byzantine-resilient substrate presented here.

Code and Data Availability

Three Monte Carlo simulator implementations supporting the empirical-validation sections (Sections VII–IX) are released at <https://github.com/ok-research/>

bft-uav-swarm-validation under the MIT licence; each section’s Reproducibility paragraph identifies the corresponding subdirectory and tag. Physics-grounded experiments in a Gazebo/ROS2 simulation environment remain deferred to a forthcoming track and will be released alongside that follow-on work. Sentinel-2 imagery used for Terrain-Referenced Navigation is publicly available through the Copernicus Open Access Hub (<https://scihub.copernicus.eu/>) under the Copernicus Data Space open-access licence; no custom dataset is generated by the present code release.

XI. CONCLUSION

We presented a Byzantine-fault-tolerant hierarchical navigation architecture for GNSS-denied UAV swarms that integrates, within a continuous-state flight-grade factor-graph estimator, six previously disjoint capabilities: BFT consensus, dynamic anchor/follower hierarchy, reputation-weighted multi-hop position relay, sub-10-second automatic failover by design, UWB-based GNSS-spoofing detection, and load balancing under progressive attrition. A systematic review of approximately 90 publications from 2020–2026 indicates that, to our knowledge, no prior published system has combined more than four of these pillars, and none has done so in a continuous-state flight-grade estimator. The central primitive, visual Proof of Observation, ties trust to terrain-anchored evidence (physically bound to terrain geometry, illumination, and the learned extractor; not forgeable by a vehicle that has not observed the location under the threat model adopted here) and integrates naturally with factor-graph state estimation via asymmetric reputation-weighted noise. Theoretical analysis exposes an explicit trade-off between quorum size and scalable Byzantine tolerance, gives a closed-form worst-case upper bound on the expected time to naïve-Byzantine exclusion ($T_{\text{exclusion}} \lesssim 215$ s under conservative parameters), and quantifies multi-hop error propagation, predicting that multi-anchor reputation-weighted fusion sustains CEP <100 m up to three hops at the reference operating point with the chosen base noise. A complete simulation methodology with pre-registered analytical predictions and pass criteria is reported as Section VI; its empirical execution in a Gazebo/ROS2 framework, together with hardware-in-the-loop validation, is the immediate next milestone.

ACKNOWLEDGEMENTS

The methods described in Sections IV-B through IV-G are the subject of pending patent applications filed with the German Patent and Trademark Office (DPMA) in April 2026, under Aktenzeichen 10 2026 001 712.2 and a related application. This disclosure is made for academic purposes and does not transfer any rights in the subject matter to third parties.

The simulator implementations in Sections VII–IX were developed with LLM-assisted code generation; all numerical results were obtained by deterministic re-execution of the released code.

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